GLOBAL STABLE SOLUTIONS OF THE ONE-PHASE BERNOULLI PROBLEM IN DIMENSION SIX

ABSTRACT. We prove that every global classical stable solution of the one-phase Bernoulli problem in $\mathbb{R}^6$ is one-dimensional. Excluding nonflat smooth stable cones in dimensions five and six using OpenAI's computer-verified algebraic inequality allows us to apply the blow-down theorem of Fernández-Real and Serra. For a hypothetical nonflat solution, this gives mean-square gradient closeness to two-sided linear profiles on large balls. Our local gradient-defect estimate converts the averaged error into a supremum bound for the defect. The direction-independent length defect then vanishes at each fixed point, giving $|Du| = 1$ in the positive phase. Harmonicity then gives $D^2u = 0$. We also obtain local curvature bounds and axial symmetry for finite-index solutions.

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CONTENTS

1. Introduction 2
1.1. Research context 2
1.2. The gradient defect 3
1.3. Applications of rigidity 4
1.4. Proof overview 5
2. Stability on a spherical link 5
2.1. The flux inequality on a five-dimensional link 7
2.2. The five-dimensional cone 9
3. A scalar defect for the gradient 11
4. Stability and an iteration 13
4.1. Energy and Sobolev inequalities 13
4.2. Raising the power of the defect 14
5. Blow-down and classification 16
6. Local curvature and finite index 18
6.1. Bounds on balls centered at the free boundary 18
6.2. Finite-index solutions 22
7. Hessian energy and neck exclusion 23
Appendix A. Scalar and inner stability 26
Appendix B. Weighted local curvature estimate 28
References 29

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1. Introduction

Let $u \geq 0$ be continuous in an open set $A \subset \mathbb{R}^n$. Write $\Omega = \{u > 0\}$ and $\Gamma = \partial\Omega \cap A$. The one-phase Bernoulli problem is

$$\Delta u = 0 \quad \text{in } \Omega, \qquad u = 0, \quad |Du| = 1 \quad \text{on } \Gamma. \tag{1.1}$$

For a classical solution, Γ is smooth, Ω lies locally on one side, and u is smooth up to Γ from Ω. Since $u = 0$ on Γ, its tangential derivatives vanish. Its inward derivative is nonnegative by positivity and equals one by (1.1). Thus the positive-side trace of Du is the inward unit normal N. We write $\nu = -N$ for the outward normal and $H = \operatorname{div}_\Gamma N$ for the sum of the principal curvatures with the inward convention.

For $\phi \in C_c^\infty(A)$, integration by parts on the positive phase gives

$$\int_A u\,\partial_i\phi = \int_\Omega u\,\partial_i\phi = -\int_\Omega (\partial_i u)\phi,$$

since the boundary term contains the zero trace of u. The ambient weak gradient is therefore Du in Ω and zero elsewhere. Smoothness up to Γ makes it locally bounded, so u is locally Lipschitz. A global solution has $A = \mathbb{R}^n$.

On an open set D compactly contained in A, the Alt–Caffarelli energy is $J(u; D) = \int_D (|Du|^2 + \mathbf{1}_\Omega)$ [1]. Take a smooth family of diffeomorphisms Φ_t with $\Phi_0 = \mathrm{id}$, equal to the identity outside a fixed compact subset of D. Stability means that the second derivative at zero of $J(u \circ \Phi_t^{-1}; D)$ is nonnegative for every such D and Φ_t. For a classical solution this gives

$$Q(\phi) := \int_\Omega |D\phi|^2 - \int_\Gamma H\phi^2 \geq 0 \tag{1.2}$$

for $\phi \in C_c^\infty(A)$ [4, Appendix A, (A.1)]. Appendix A gives the passage to signed tests and the converse. Boundary integrals use surface measure and tensor norms are Euclidean.

A logarithmic cutoff argument gives one-dimensionality for entire classical stable solutions in the plane [12, Theorem 3.1]. Chan, Fernández-Real, Figalli, and Serra proved rigidity in dimension three [4, Theorem 1.5], and Fernández-Real and Serra proved it in dimension four [9, Theorem 1.1].

Theorem 1.1 (Global classical stable rigidity)**.** *Let u be a global classical stable solution of the one-phase Bernoulli problem in $\mathbb{R}^6$. Then $D^2u = 0$ in Ω. If Γ is empty, u is a nonnegative constant. Otherwise there is a unit vector e such that u has one of the following forms, where $t_+ = \max(t, 0)$:*

$$u(x) = (e \cdot x - a)_+ \qquad \text{for some } a \in \mathbb{R},$$
$$u(x) = (e \cdot x - b)_+ + (a - e \cdot x)_+ \qquad \text{for some } a < b.$$

In Figure 1, the gradients on the positive pieces are e for the half-space and $-e, e$ for the two sides of the strip. These affine pieces are harmonic and have unit gradient. The boundary planes have $H = 0$, so $Q(\phi) = \int_\Omega |D\phi|^2 \geq 0$ and both profiles are stable by Appendix A.

1.1. Research context.

Weiss's monotonicity formula relates singularities of minimizing free boundaries to nonflat one-homogeneous minimizing cones [14]. Caffarelli, Jerison, and Kenig proved full regularity for minimizing free boundaries in dimension three [3]. Jerison and Savin excluded nonflat smooth stable cones through dimension four and obtained regularity for minimizers in that range [11, Theorem 1.1 and Corollary 1.2]. The bi-orthogonally symmetric cones in [10, Theorem 1.1] are unstable in dimensions three through six and strictly stable from dimension seven.

The first dimension admitting a nonflat one-homogeneous global minimizer is seven [13, Theorem 1.1]. The minimizing cone U_0 of De Silva and Jerison has positive phase

$$\mathcal{C} = \{(x', x_7) : |x_7| < c|x'|\}, \qquad c > 0,$$

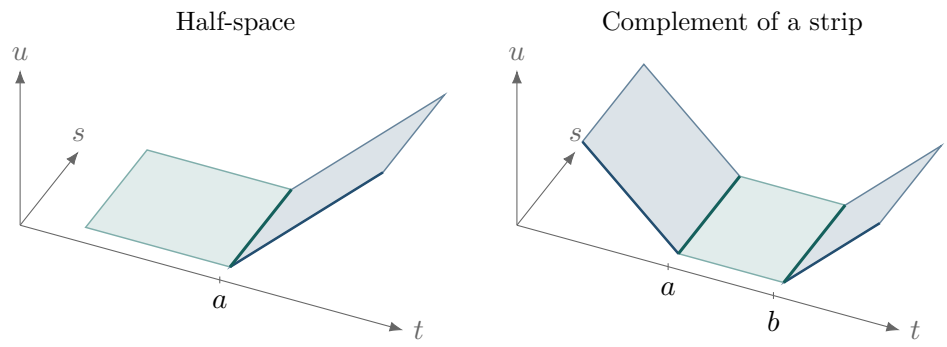


FIGURE 1. Graphs of the two nonconstant profiles over a slice with $t = e \cdot x$ and transverse coordinate s. The height u is constant along s. The pale teal patches mark the zero phase, and the dark teal edges mark the free-boundary traces.

and its free boundary is smooth away from the vertex [5, Theorem 1.1]. The cone $\mathcal{C}$ is connected because every point can be joined vertically to the connected punctured hyperplane $\{x_7 = 0,\ x' \neq 0\}$.

For this cone, De Silva, Jerison, and Shahgholian construct a classical global minimizer $U \geq U_0$ with analytic free boundary and $R^{-1}U(R\,\cdot) \to U_0$ locally uniformly as $R \to \infty$ [6, Theorem 1.1 and Lemma 3.1]. Minimality gives stability since each compactly supported deformation is an admissible competitor in a ball containing its support. If $D^2U = 0$ in $\{U > 0\}$, then U is affine on the connected subset $\mathcal{C} \subset \{U > 0\}$, say $U(x) = a \cdot x + b$ there. For $p = (p', 0)$ with $p' \neq 0$, both Rp and $-Rp$ lie in $\mathcal{C}$ for $R > 0$. Thus the blow-down gives

$$U_0(p) + U_0(-p) = \lim_{R\to\infty} \frac{U(Rp) + U(-Rp)}{R} = \lim_{R\to\infty} \frac{2b}{R} = 0.$$

This contradicts $U_0(\pm p) > 0$ and proves sharpness of Theorem 1.1 in dimension seven. See [7] for a survey of the minimizing and stable theories.

1.2. **The gradient defect.** Writing $t = e \cdot x$, the strip profile in Theorem 1.1 rescales to

$$\frac{u(Rx)}{R} = (t - b/R)_+ + (a/R - t)_+.$$

The zero strip has width $(b-a)/R$. Since the positive-part map is 1-Lipschitz, the rescaled profile differs from $t_+ + (-t)_+ = |t|$ by at most $(|a| + |b|)/R$. The limiting slopes e and $-e$ motivate a defect that vanishes at both. With $P_e p = p - (p \cdot e)e$, set

$$h_e = \left(|P_e Du|^2 + \frac{(1 - |Du|^2)^2}{30} \right)^{1/2} \quad \text{in } \Omega.$$

The first term under the square root vanishes when Du is a multiple of e, and the second vanishes when $|Du| = 1$. Thus $h_e = 0$ precisely at $Du = \pm e$. The next estimate converts small mean-square defect on a ball into pointwise control on the ball of half the radius.

Theorem 1.2 (Local gradient-defect estimate)**.** *There are dimensional constants $C, \varepsilon_0 > 0$ with the following property. Let u be a classical stable solution in $B_R(x_0) \subset \mathbb{R}^6$, with $|Du| \leq 1$ in its positive phase in this ball. For a unit vector e put*

$$E_e = \left(R^{-6} \int_{\Omega \cap B_R(x_0)} h_e^2 \right)^{1/2}.$$

If $E_e \le \varepsilon_0$, then

$$\sup_{\Omega \cap B_{R/2}(x_0)} h_e \le C E_e.$$

1.3. **Applications of rigidity.** To apply Theorem 1.1 in lower dimensions, extend the solution constantly in the extra coordinates. This adds only zero principal curvatures to the free boundary. Slicing an ambient test in those coordinates recovers the original stability form and adds a nonnegative Dirichlet term. The product argument is given in Section 5.

Corollary 1.3 (Rigidity in dimensions at most five)**.** *Let $1 \le n \le 5$ and let u be a global classical stable Bernoulli solution in $\mathbb{R}^n$. Then $D^2 u = 0$ in Ω. If Γ is empty, u is a nonnegative constant. If $\Gamma \neq \varnothing$, u has one of the two forms in Theorem 1.1, with e a unit vector in $\mathbb{R}^n$.*

Local Hessian and curvature bounds were proved in dimension three [4, Corollary 1.7] and dimension four [9, Corollary 1.3]. In Section 6 we rescale at points of large Hessian to obtain a global limit with Hessian of norm one at a point. If the limit is classical, rigidity forces its Hessian to vanish. If two limiting sheets touch, the limit is an absolute-value profile, affine on each positive side. The curvature bound follows from the Hessian bound because $D^2 u(X, Y) = \langle D_X N, Y \rangle$ for tangent vectors X, Y on Γ.

Corollary 1.4 (Local Hessian and free-boundary curvature estimates)**.** *Let $1 \le n \le 6$ and let u be a classical stable solution in $B_{2R}(x_0) \subset \mathbb{R}^n$, with $x_0 \in \Gamma$. Then*

$$\sup_{\Omega \cap B_R(x_0)} |D^2 u| \le \frac{C_n}{R}, \qquad \sup_{\Gamma \cap B_R(x_0)} |\mathrm{II}_\Gamma| \le \frac{C_n}{R}. \tag{1.3}$$

Here C_n depends only on the dimension and II_Γ is the second fundamental form. For $n = 1$, $u'' = 0$ on the positive intervals and the second inequality is vacuous.

The scalar Morse index is the supremum of the dimensions of subspaces of $C_c^\infty(\mathbb{R}^n)$ on which Q is negative definite.

Finite index implies stability outside some ball. Otherwise we could choose k tests with $Q(\phi_i) < 0$, each supported outside a ball containing all the earlier supports. The supports are disjoint in both terms of (1.2), so

$$Q\left(\sum_{i=1}^k a_i \phi_i\right) = \sum_{i=1}^k a_i^2 Q(\phi_i) < 0 \qquad \text{for } (a_1, \dots, a_k) \neq 0.$$

This produces negative subspaces of arbitrarily large dimension.

Basulto and Kamburov classified entire classical solutions in the plane with finite Morse index and connected positive phase [2, Theorem 1.1]. Fernández-Real, Florit-Simon, and Serra proved axial symmetry in dimension three and obtained a higher-dimensional classification under rigidity hypotheses for entire stable solutions and smooth stable cones [8, Theorems 1.1 and 1.2]. Section 6 checks both hypotheses through dimension six.

Corollary 1.5 (Finite-index axial symmetry)**.** *Let $3 \le n \le 6$. Every entire classical Bernoulli solution of finite scalar Morse index is axially symmetric about a line. If u cannot be written as a function of one linear coordinate, then, after a translation, rotation, and Bernoulli scaling, it has one of the following forms.*

1. *The exterior-ball solution. It is zero on $\{|x| \le 1\}$ and equals $(1 - |x|^{2-n})/(n-2)$ on $\{|x| > 1\}$.*
2. *A solution $u(x', x_n) = U(r, x_n)$, $r = |x'|$, that is even in x_n, with $u_{x_n} > 0$ in $\Omega \cap \{x_n > 0\}$ and $U_r < 0$ in $\Omega \cap \{r > 0\}$. Its zero phase is*

$$\{(x', x_n) : r \ge 1,\ |x_n| \le g(r)\}.$$

The function g is continuous on $[1,\infty)$, smooth on $(1,\infty)$, and satisfies $g(1)=0$, $g'(r)>0$ for $r>1$, and $\lim_{r\to\infty} g(r)\in(0,\infty)$.

1.4. **Proof overview.** Let n_* be the first dimension $n\geq 2$ admitting a nonflat one-homogeneous stable solution with a free boundary through the origin that is classical away from the origin [9, Definition 4.1]. Section 2 reduces stability for a cone to an inequality on its spherical cross-section and uses [13, Proposition 4.1] to exclude these cones in dimensions five and six. Together with [11, Theorem 1.1], this gives $6<n_*$.

For a solution with nonzero Hessian, center a free-boundary point at the origin and rescale to arrange $\sup_{\Omega\cap B_1}|D^2u|\geq 1$ as in Section 5. The blow-down estimate [9, Proposition 4.12] then gives directions e_R and profiles $V_R(x)=|e_R\cdot x|$ such that

$$R^{-6}\int_{B_R}|Du-DV_R|^2\longrightarrow 0\qquad\text{as } R\to\infty.$$

The absolute-value profile has a positive phase on both sides of its ridge, so it falls outside the classical cone exclusion. For $q=DV_R=\pm e_R$ almost everywhere, $P_{e_R}q=0$, so $|P_{e_R}Du|\leq|Du-q|$. The global bound $|Du|\leq 1$ [4, Lemma 3.2] gives $1-|Du|^2=(q-Du)\cdot(q+Du)\leq 2|Du-q|$. Thus $h_{e_R}^2\leq(1+4/30)|Du-DV_R|^2$, giving $E_{e_R}\to 0$ on B_R.

For the local estimate, Section 3 shows that the regularized defect $h_\delta=(h_e^2+\delta^2)^{1/2}$ satisfies $h_\delta\Delta h_\delta\geq|D^2u|^2/75$. Set $b=(1+|Du|^2)/2$ and $t=h_\delta/b$. On Γ, $\partial_\nu b=Hb$ and $\partial_\nu t\leq 0$, so $G=bF(t)$ satisfies

$$\partial_\nu G-HG=bF'(t)\partial_\nu t\leq 0\qquad\text{when } F'\geq 0.$$

For $F\geq 0$, this sign makes the boundary term nonpositive when stability is applied to a cutoff of G and integrated by parts. Section 4 continues powers F by their tangent lines above a fixed threshold, keeping $(tF'-F)/F'$ small so that $F'\Delta h_\delta$ absorbs the bulk error $-(tF'-F)|D^2u|^2$. Iterating the resulting estimates gives the supremum bound.

Applying Theorem 1.2 to the blow-down errors gives $\sup_{\Omega\cap B_{R/2}}h_{e_R}\to 0$. At every fixed point in Ω, the direction-independent bound $1-|Du|^2\leq\sqrt{30}\,h_{e_R}$ then gives $|Du|=1$. Since u is harmonic, $\Delta|Du|^2=2|D^2u|^2$, which contradicts the nonzero Hessian. Section 5 completes the classification of the resulting affine pieces.

Proposition 7.1 in Section 7 bounds the Hessian energy when the height excess is small. Appendix B gives the weighted local curvature estimate.

2. Stability on a spherical link

For the blow-down theorem of [9, Proposition 4.12] we need $6<n_*$, where n_* is defined in Section 1.4. Jerison and Savin exclude nonflat stable cones that are classical off the origin in dimensions two through four [11, Theorem 1.1]. We treat dimensions five and six by comparing the stability inequality on the spherical cross-section with the flux of a vector field there.

Let u be a nonzero one-homogeneous Bernoulli solution in $\mathbb{R}^n$, $n\in\{5,6\}$, that is classical and stable away from the origin. Write $u(r\theta)=rg(\theta)$ for $r>0$, $\theta\in\mathbb{S}^m$, and $m=n-1$. Let D be a connected component of $\{g>0\}$. The radial direction is tangent to the conical free boundary and normal to the unit sphere. Thus the intersection is transverse. Classicality gives a smooth boundary for D and smoothness of g up to that boundary from inside. All link derivatives use the round metric. Write N for the inward unit normal to ∂D within the sphere.

For a Euclidean displacement $V=\alpha\theta+Y$ with Y tangent to the sphere, $D_Vr=\alpha$ and $D_V\theta=Y/r$. Hence $D_Vu=\alpha g+\langle\nabla g,Y\rangle$ and $Du=g\theta+\nabla g$. To differentiate again, use the ambient derivative of a tangent field Z on the unit sphere: $D_YZ=\nabla_YZ-\langle Y,Z\rangle\theta$. The radial part of $D_Y\nabla g$ cancels $(Yg)\theta$ from $D_Y(g\theta)$, leaving $\nabla_Y\nabla g+gY$. Put $T=\nabla^2g+gI$, $B=\nabla T$, and $w=|T|$, identifying symmetric bilinear forms with endomorphisms by the metric. Then $D_V(Du)=r^{-1}TY$.

Taking the Euclidean trace gives $0 = r\Delta_{\mathbb{R}^n} u = \Delta g + mg$. On ∂D, $g = 0$ and the Bernoulli condition gives $|\nabla g| = 1$. The tangential derivatives along ∂D vanish and g is positive on the inward side. Thus

$$\Delta g = -mg \quad \text{in } D, \qquad g = 0, \quad \nabla g = N \quad \text{on } \partial D. \tag{2.1}$$

The boundary is nonempty. Otherwise D would be the whole sphere, where integrating $\Delta g = -mg$ would contradict $g > 0$. The Hessian calculation above, in a frame with one radial and m spherical directions, reads

$$D^2 u(r\theta) = r^{-1} \operatorname{diag}(0, T(\theta)). \tag{2.2}$$

Equation (2.1) gives $\operatorname{tr} T = 0$. Use an orthonormal frame at the point of computation, with indices from 1 to m and repeated indices summed. Our curvature convention is $R(X,Y)Z = \langle Y, Z\rangle X - \langle X, Z\rangle Y$ on the unit sphere. The commutator of covariant derivatives acts on each covariant slot by minus the action of R on vectors. For $g_k = \nabla_k g$ and $g_{jk} = \nabla_j g_k$, this gives

$$\nabla_i g_{jk} - \nabla_j g_{ik} = -g_i \delta_{jk} + g_j \delta_{ik}.$$

The derivatives of the gI term cancel the right-hand side, so $B_{ijk} = \nabla_i T_{jk}$ is symmetric in i, j. Since T is symmetric, B is fully symmetric. Also $\sum_j B_{ijj} = \nabla_i \operatorname{tr} T = 0$, so the trace on any two slots vanishes. For T, the commutator in directions ℓ, i contributes $-T(R(e_\ell, e_i)e_j, e_k) - T(e_j, R(e_\ell, e_i)e_k)$. Substituting the curvature formula gives

$$\nabla_\ell B_{ijk} - \nabla_i B_{\ell jk} = \delta_{\ell j} T_{ik} + \delta_{\ell k} T_{ij} - \delta_{ij} T_{\ell k} - \delta_{ik} T_{\ell j}. \tag{2.3}$$

To contract this identity, set $j = \ell$ and sum over ℓ. On the left, $\sum_\ell \nabla_\ell B_{i\ell k} = \Delta T_{ik}$ by symmetry of B, while $\nabla_i \sum_\ell B_{\ell\ell k} = 0$. The right side is $mT_{ik} + T_{ik} - T_{ik} - \delta_{ik} \operatorname{tr} T$. This proves $\Delta T = mT$ for the rough Laplacian, the trace of the second covariant derivative. The product rule gives $\Delta(w^2) = 2\langle T, \Delta T\rangle + 2|B|^2$. Where $w > 0$, Cauchy–Schwarz applied to $w\nabla_\ell w = T_{ij} B_{\ell ij}$ bounds $|\nabla w|$. Thus

$$\Delta w^2 = 2mw^2 + 2|B|^2, \qquad |\nabla w| \le |B| \quad (w > 0). \tag{2.4}$$

Set $\sigma = g^2 + |\nabla g|^2$. Since $\nabla_i \sigma = 2gg_i + 2g_{ij} g_j = 2T_{ij} g_j$, we have

$$\Delta\sigma = 2(\nabla_i T_{ij}) g_j + 2T_{ij} g_{ij} = 2T_{ij}(T_{ij} - g\delta_{ij}) = 2w^2.$$

Here $\nabla_i T_{ij} = B_{iij} = 0$ by trace-freeness of B. On ∂D, $\sigma = 1$ by (2.1). If $T \not\equiv 0$ on D, the strong maximum principle gives $\sigma < 1$ in D, and the Hopf lemma gives $\partial_N \sigma < 0$ at each boundary point. Tangential differentiation of $\sigma = 1$ shows that TN is parallel to N. For X tangent to ∂D, differentiating $\nabla g = N$ along ∂D gives $TX = \nabla_X N$, because $g = 0$ there. Thus the tangential block of T is the second fundamental form with respect to N. Write H_D for its trace. Since T has trace zero,

$$TN = -H_D N, \qquad \partial_N \sigma = -2H_D.$$

Consequently $w \ge H_D > 0$ on ∂D. The boundary flux uses the entries of B with an odd number of normal slots. For tangent vectors X, Y to ∂D, differentiate $T(N, Y) = 0$ along X. Using $\nabla_X N = TX$ and $\langle \nabla_X Y, N\rangle = -\langle Y, \nabla_X N\rangle = -T(X, Y)$, we get

$$0 = B(X, N, Y) + T(TX, Y) - T(N, N)T(X, Y).$$

By symmetry of B, this gives the first of the two identities

$$B(N, X, Y) = T(N, N)T(X, Y) - (T^2)(X, Y), \qquad B(N, N, N) = w^2.$$

Indeed, summing the first over an orthonormal tangent basis of ∂D gives $-T(N,N)^2 - \sum_{\alpha,\beta} T_{\alpha\beta}^2 = -w^2$. We used $\operatorname{tr} T = 0$ and the vanishing of the mixed entries of T. The second identity follows by taking the full trace of $B(N, \cdot, \cdot)$.

Let φ be smooth on $\overline{D}$, with support away from any singular link points when using this calculation on a regular portion of a link. At each regular boundary point of its support, a smooth boundary chart contains only the one positive side belonging to D. Finitely many such

charts cover the compact boundary part of the support. Extend φ across these charts and use a partition of unity with support in the charts and the interior. The resulting ambient angular test agrees with φ on $\overline{D}$ and vanishes near every other positive component and every singular point in the ambient link. For $\zeta \in C_c^\infty((0,\infty))$, the product test $\zeta(r)\varphi(\theta)$ using this extension has support in a compact annulus on the regular part of the ambient cone.

On the conical side the tangent space is spanned by θ and the tangents to ∂D. The inward normal is $N(r\theta) = N(\theta)$. Its radial derivative is zero, and in each unit tangent direction Y along ∂D its Euclidean derivative is $r^{-1}\nabla_Y N$. There is no radial term in this derivative because $\langle Y, N\rangle = 0$. Thus the Euclidean mean curvature is $H(r\theta) = r^{-1}H_D(\theta)$. The angular tangent directions scale by r, so the bulk measure is $r^{n-1}\,dr\,d\theta$ and the side measure is $r^{n-2}\,dr\,dS_\theta$. For the product test,

$$|D(\zeta\varphi)|^2 = |\zeta'|^2\varphi^2 + r^{-2}\zeta^2|\nabla\varphi|^2.$$

Substituting into (1.2) gives a common radial factor $\int_0^\infty r^{n-3}\zeta^2\,dr$ in the boundary and angular terms. Dividing by it for $\zeta \neq 0$ yields

$$\int_{\partial D} H_D\varphi^2 \le \int_D |\nabla\varphi|^2 + \frac{\int_0^\infty r^{n-1}|\zeta'|^2\,dr}{\int_0^\infty r^{n-3}\zeta^2\,dr}\int_D \varphi^2.$$

Put $\mu = (n-2)/2$ and choose $\zeta(r) = r^{-\mu}\chi(\log r/L)$, where χ is a nonzero smooth compactly supported function. Differentiating gives $\zeta'(r) = r^{-\mu-1}(L^{-1}\chi'(\log r/L) - \mu\chi(\log r/L))$. In both integrals the remaining radial factor is $r^{-1}dr = dt$ for $t = \log r$. The denominator is $\int_{\mathbb{R}} \chi(t/L)^2\,dt$ and the numerator is

$$\int_{\mathbb{R}} \left(L^{-1}\chi'(t/L) - \mu\chi(t/L)\right)^2\,dt.$$

The cross term vanishes by $\int_{\mathbb{R}} \chi'\chi = 0$, so the quotient is $\mu^2 + L^{-2}\int|\chi'|^2/\int\chi^2$. As $L \to \infty$ this gives

$$\int_{\partial D} H_D\varphi^2 \le \int_D (|\nabla\varphi|^2 + \tfrac{(n-2)^2}{4}\varphi^2). \tag{2.5}$$

The angular support stays fixed in this limit. For a suspended link, we apply (2.5) to each fixed test away from the tips before removing its angular cutoffs.

2.1. **The flux inequality on a five-dimensional link.** In dimension six, (2.5) bounds the boundary integral of $H_D\varphi^2$ by the integral of $|\nabla\varphi|^2 + 4\varphi^2$. We look for a test h_{s} and a vector field Z whose divergence controls this same bulk expression with a strict positive term, while its outward flux is bounded above by $H_D h_{\mathrm{s}}^2$. Integrating would then contradict stability. The pointwise comparisons of [13, Propositions 5.2 and 5.4] come from an algebraic certificate [13, Proposition 4.1]. Their proofs use the local tensor identities above. We check how those identities enter the comparisons before applying stability to the cutoff tests.

On the part of a link in $\mathbb{S}^5$ where $w > 0$, set $a = 29/50$. In a spherical orthonormal frame put

$$A = T/w, \qquad X = B/w, \qquad p = \nabla g, \qquad q_l = A_{ij}X_{lij}.$$

The matrix A is symmetric and trace free with $|A| = 1$. The tensor X is symmetric and trace free, and $g^2 + |p|^2 \le 1$. These are the admissibility conditions in [13, Proposition 4.1]. Differentiating $w^2 = T_{ij}T_{ij}$ and using the quotient rule for A gives the first two identities below. The last two use $p = \nabla g$ and $T = \nabla^2 g + gI$.

$$\nabla_l w = wq_l, \qquad \nabla_l A_{ij} = X_{lij} - q_l A_{ij}, \qquad \nabla_l p_i = wA_{li} - g\delta_{li}, \qquad \nabla_l g = p_l.$$

In a frame with vanishing connection coefficients at the point, these are the differentiation rules D_l of the certificate at $z = 1$. The product rule and $\nabla_l\delta_{ij} = 0$ extend this identification to the coefficient polynomials in A, p, g, δ. We differentiate w and X separately.

Use the coefficient polynomials f, U, K, M, P of that proposition, given by Table I with the decoding in [13, Section 6.1]. With $C_{lijk} = K_{lijk} - K_{iljk}$ and $G_{lijk} = \delta_{li}U_{jk} + C_{lijk}$, the fields are

$$h_{\mathrm{s}} = w^a\sqrt{f}, \qquad Z_l = w^{2a}(G_{lijk}X_{ijk} + wM_l + P_l). \tag{2.6}$$

Orthogonal equivariance makes h_{s} and Z independent of the chosen frame. The certificate gives $f \geq c_* > 0$ on the admissible data. To see how the interior comparison applies, the only fourth derivatives of g in $\operatorname{div} Z$ enter through $\nabla_l X_{ijk} = w^{-1}\nabla_l B_{ijk} - q_l X_{ijk}$. The $\delta_{li}U_{jk}$ part of G contracts the first term to $U_{jk}\Delta T_{jk}/w = 5U_{jk}A_{jk}$. The C_{lijk} part is skew in l, i. Thus (2.3) gives

$$\begin{aligned} w^{-1}C_{lijk}\nabla_l B_{ijk} &= \tfrac{1}{2w}C_{lijk}(\nabla_l B_{ijk} - \nabla_i B_{ljk}) \\ &= C_{lijk}(\delta_{lj}A_{ik} + \delta_{lk}A_{ij}). \end{aligned}$$

In the second line, exchanging l, i in the last two terms of the commutator makes them equal to the first two after contraction. Thus no fourth derivatives remain. The factors of w to differentiate in the three summands of Z are w^{2a-1} in $w^{2a}X = w^{2a-1}B$, w^{2a+1} in the M term, and w^{2a} in the P term. Using $\nabla_l w = wq_l$ gives the coefficients $2a-1$, $2a+1$, and $2a$ of q_l in the divergence identity of [13, Proposition 5.2].

Put $\mathcal{F}_l = D_l f + 2aq_l f$. Differentiating h_{s} gives $\nabla_l h_{\mathrm{s}} = w^a\mathcal{F}_l/(2\sqrt{f})$. The interior certificate at $z = 1$ says that

$$w^{-2a}\operatorname{div} Z - 4f + f|\ell|^2 - \mathcal{F}\cdot\ell \geq c_*(1 + |\ell|^2) \qquad (\ell \in \mathbb{R}^5).$$

For $\ell = \mathcal{F}/(2f)$, the last two terms on the left combine to $-|\mathcal{F}|^2/(4f) = -w^{-2a}|\nabla h_{\mathrm{s}}|^2$. Multiplying by w^{2a} yields the interior comparison.

On a regular side boundary, $g = 0$, $p = N$, and $Ap = \lambda p$ with $|p| = 1$ and $\lambda = -H_D/w < 0$. Set $S_0 = \lambda A - A^2$. The boundary identities above give $B(N, Y_1, Y_2)/w^2 = S_0(Y_1, Y_2)$ for Y_1, Y_2 tangent to ∂D, and $B(N, N, N)/w^2 = 1$. Reflection in $p^\perp$ fixes A and reverses p. Equivariance and evenness in p leave G invariant under this reflection, so $p_l G_{lijk}$ has only components with an odd number of normal slots. The boundary identities therefore determine its contraction with B/w^2. The remaining vector term P vanishes at $g = 0$, so the signed boundary comparison applies with inward normal N [13, Lemma 5.3 and Proposition 5.4].

We have proved the comparisons

$$\operatorname{div} Z \geq |\nabla h_{\mathrm{s}}|^2 + 4h_{\mathrm{s}}^2 + cw^{2a}, \tag{2.7}$$

$$N\cdot Z + H_D h_{\mathrm{s}}^2 \geq 0 \quad \text{on the smooth side of the link.} \tag{2.8}$$

For the size bounds, the coefficient polynomials and their first derivatives in (A, p, g) are bounded on the compact normalized coefficient domain. The differentiation rules above and $|q| \leq |X|$ give $|\mathcal{F}| \leq C(1 + w + |X|)$. Using $f \geq c_*$, the formula for ∇h_{s}, and (2.6) gives

$$|h_{\mathrm{s}}| \leq Cw^a, \quad |\nabla h_{\mathrm{s}}| \leq Cw^a(1 + w + |B|/w), \quad |Z| \leq Cw^{2a}(1 + w + |B|/w). \tag{2.9}$$

Let $0 \leq \xi \leq 1$ be smooth up to the regular boundary and vanish near the zeros of w and any tips. All products below involving ξ or $\nabla\xi$ are extended by zero there. Integrate (2.7) against ξ^2. The divergence theorem for $\xi^2 Z$ uses outward normal $-N$. In the suspended case we may truncate the domain at $\tau = \varepsilon, \pi - \varepsilon$ inside the regions where $\xi = 0$, so the new boundary faces contribute zero. Using $-N\cdot Z \leq H_D h_{\mathrm{s}}^2$ gives

$$\begin{aligned} \int_D \xi^2(|\nabla h_{\mathrm{s}}|^2 + 4h_{\mathrm{s}}^2 + cw^{2a}) &\leq \int_D \xi^2 \operatorname{div} Z \\ &= -\int_{\partial D}\xi^2 N\cdot Z - 2\int_D \xi Z\cdot\nabla\xi \leq \int_{\partial D} H_D(\xi h_{\mathrm{s}})^2 - 2\int_D \xi Z\cdot\nabla\xi. \end{aligned}$$

Apply (2.5) to ξh_{s} with $n = 6$, so its mass coefficient is 4. The expansion

$$|\nabla(\xi h_{\mathrm{s}})|^2 = \xi^2|\nabla h_{\mathrm{s}}|^2 + h_{\mathrm{s}}^2|\nabla\xi|^2 + 2\xi h_{\mathrm{s}}\nabla h_{\mathrm{s}}\cdot\nabla\xi$$

allows us to cancel the common Dirichlet and mass terms before taking any limits. Thus

$$c\int_D \xi^2 w^{2a} \le \int_D \left(h_{\mathrm{s}}^2|\nabla\xi|^2 + 2\xi(h_{\mathrm{s}}\nabla h_{\mathrm{s}} - Z)\cdot\nabla\xi\right). \tag{2.10}$$

Write $\mathcal{E}_\xi$ for the integrand on the right. Both dimensions use the same estimate from (2.9):

$$|\mathcal{E}_\xi| \le Cw^{2a}|\nabla\xi|(|\nabla\xi| + 1 + w + |B|/w) \quad \text{on } \{w>0\}.$$

Differentiation of the cutoff near $w=0$ costs a factor $|B|/w$. The weight w^{2a} in (2.10) therefore leads to the integral of $w^{2a-2}|B|^2$. To bound it on an original smooth link in either dimension, put $W_\delta = w^2+\delta$. This is smooth even where $w=0$. Equation (2.4) gives $\Delta W_\delta \ge 2|B|^2$. Also $\nabla_l W_\delta = 2T_{ij}B_{lij}$, so $|\nabla W_\delta|^2 \le 4w^2|B|^2 \le 4W_\delta|B|^2$. Because $a-1<0$, we use the upper bound for the gradient term in the chain rule. The lower bound is positive since $a>1/2$. Thus

$$\Delta W_\delta^a = aW_\delta^{a-1}\Delta W_\delta + a(a-1)W_\delta^{a-2}|\nabla W_\delta|^2 \ge 2a(2a-1)W_\delta^{a-1}|B|^2.$$

If $T\not\equiv 0$, then $w>0$ on ∂D. The zero set is therefore compact in D. Choose $0\le\psi\le 1$ in $C_c^\infty(D)$ equal to one near this set. Then

$$2a(2a-1)\int_D \psi W_\delta^{a-1}|B|^2 \le \int_D \psi\,\Delta W_\delta^a = \int_D W_\delta^a\Delta\psi \le \|W_\delta^a\|_\infty\int_D|\Delta\psi|.$$

The right side is bounded uniformly for $0<\delta\le 1$ by boundedness of w on $\overline{D}$. On $\{w>0\}$, the weight W_δ^{a-1} increases to w^{2a-2} as $\delta\downarrow 0$. Away from the neighborhood of the zero set, w is bounded away from zero on the compact closure. Monotone convergence therefore gives

$$\int_{\{w>0\}} w^{2a-2}|B|^2 < \infty. \tag{2.11}$$

In particular the function

$$M = w^{2a}(1+|B|/w)^2 \quad \text{on } \{w>0\}, \qquad M=0 \quad \text{on } \{w=0\},$$

is integrable on any original nonflat link, since $M\le 2w^{2a}+2w^{2a-2}|B|^2$ on $\{w>0\}$. Fix a smooth nondecreasing cutoff ϑ with values in $[0,1]$, equal to zero on $[0,1]$ and one on $[2,\infty)$. The cutoff $\vartheta(jw)$ is smooth across the zero set because it vanishes on a neighborhood of it. On $\{w>0\}$,

$$|\nabla\vartheta(jw)| \le C|B|/w,$$

since the derivative is supported in $1\le jw\le 2$, where $j\le 2/w$ and $|\nabla w|\le|B|$. For an original link of a cone in $\mathbb{R}^6$, use $\xi_j=\vartheta(jw)$. The error $|\mathcal{E}_{\xi_j}|$ is bounded by CM because w is bounded. The suspended link requires an additional cutoff at each tip.

2.2. **The five-dimensional cone.** Suppose a stable cone in $\mathbb{R}^5$ has a nonflat component $D_0\subset\mathbb{S}^4$, with fields g_0, T_0, B_0, w_0 as above. Extend the cone by $\widetilde{u}(x,t)=u_0(x)$ on $\mathbb{R}^5\times\mathbb{R}$. The extension is one-homogeneous and solves the classical Bernoulli problem off the axis $\{x=0\}$, because $\Delta\widetilde{u}=\Delta u_0$ and $D\widetilde{u}=(Du_0,0)$. The product boundary has the original principal curvatures and one additional zero. For an ambient test ϕ supported in a compact set off the axis, each slice $\phi(\cdot,t)$ has compact support off the vertex of u_0. The product splitting of the Dirichlet integral and Fubini give

$$Q_{\widetilde{u}}(\phi) = \int_{\mathbb{R}} Q_{u_0}(\phi(\cdot,t))\,dt + \int_{\{\widetilde{u}>0\}}|\partial_t\phi|^2 \ge 0.$$

This is the stability needed on the regular part of the suspension.

The new link is the suspension in Figure 2. In coordinates it is given by

$$\theta = (\sin\tau\, y, \cos\tau), \qquad 0<\tau<\pi, \quad y\in D_0, \qquad g(\theta)=\sin\tau\, g_0(y).$$

Write $s=\sin\tau$ and $c_\tau=\cos\tau$. If $E_1,\dots,E_4$ form an orthonormal tangent frame on the base sphere at y, then $e_\tau=(c_\tau y,-s)$ and $e_\alpha=(E_\alpha,0)$ form an orthonormal frame at θ. Indeed, the

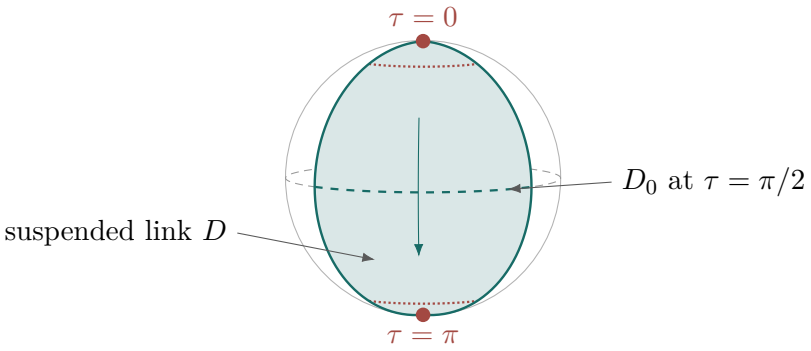


FIGURE 2. A lower-dimensional sketch of a spherical suspension. Each y in the base link runs along a meridian as τ varies. The dotted arcs mark the two latitudes where $\sin\tau = 1/j$ in the tip cutoff. The test vanishes in the caps beyond these arcs.

derivative in a base direction E_α is $(sE_\alpha, 0)$, perpendicular to e_τ. The metric is $d\tau^2 + s^2 g_{\mathbb{S}^4}$, and the four scaled directions give $dV_D = s^4\, d\tau\, dV_{D_0}$.

The ambient Hessian $D^2\widetilde{u}(\theta)$ is $D^2u_0(sy)$ with an extra zero row and column. From (2.2) and homogeneity, $w = w_0/s$. The product has the mean curvature of the original cone at sy, namely H_{D_0}/s. At unit radius this is also the link curvature $H_D > 0$. Moreover $g^2 + |\nabla g|^2 = |D\widetilde{u}(\theta)|^2 = |Du_0(y)|^2 = g_0^2 + |\nabla_0 g_0|^2 \le 1$. These facts verify the pointwise admissibility and boundary signs without applying a maximum principle at the tips.

To compute B, first note that for any one-homogeneous v, the restriction of D^3v at unit radius to three spherical tangent slots is the covariant derivative of the link tensor T. Indeed, differentiating $D^2v(Y,Z) = T(Y,Z)$ along a spherical tangent and subtracting the derivatives of the tangent fields gives ∇T. The radial parts of the ambient derivatives of the tangent fields make no contribution because D^2v vanishes in a radial slot. For $v = u_0$, differentiating $D^2u_0(ry) = r^{-1}D^2u_0(y)$ along the ray gives $D^3u_0(y)[y,\cdot,\cdot] = -D^2u_0(y)$. Consequently the third derivative vanishes when at least two slots are radial. By homogeneity $D^3u_0(sy) = s^{-2}D^3u_0(y)$, and the third derivative of the product ignores the last ambient coordinate. Evaluating it in the frame above gives

$$B_{\alpha\beta\gamma} = s^{-2}(B_0)_{\alpha\beta\gamma}, \qquad B_{\tau\alpha\beta} = -c_\tau s^{-2}(T_0)_{\alpha\beta}, \qquad B_{\tau\tau\alpha} = B_{\tau\tau\tau} = 0.$$

A single meridional slot occurs in three positions. Thus

$$|B|^2 = s^{-4}(|B_0|^2 + 3c_\tau^2 w_0^2). \tag{2.12}$$

Use $\xi_j(\tau, y) = \alpha_j(y)\beta_j(\tau)$, where $\alpha_j = \vartheta(jw_0)$ and $\beta_j = \vartheta(js)$. For each j, the support is separated from the tips by $s \ge 1/j$. In (2.5) the radial support has $r \ge r_{\min} > 0$, so on the product support $|x| = rs \ge r_{\min}/j$. This keeps each Euclidean test off the singular axis for its fixed radial cutoff.

The metric gives

$$|\nabla\xi_j|^2 = \alpha_j^2|\beta_j'|^2 + s^{-2}\beta_j^2|\nabla_0\alpha_j|^2.$$

On the support of β_j', $j \le 2/s$, so $|\beta_j'| = |j\vartheta'(js)c_\tau| \le C/s$. For $w_0 > 0$, set $b_0 = |B_0|/w_0$. The bound for the original cutoff gives $|\nabla_0\alpha_j| \le Cb_0$, and hence $|\nabla\xi_j| \le C(1+b_0)/s$. Equation (2.12) gives $|B|/w = s^{-1}\sqrt{b_0^2 + 3c_\tau^2}$. Since $w = w_0/s$ and w_0 is bounded, the factor $1 + w + |B|/w$ is also bounded by $C(1+b_0)/s$. Multiplying the error estimate by the volume factor gives

$$s^4|\mathcal{E}_{\xi_j}| \le Cs^{4-2a-2}w_0^{2a}(1+b_0)^2 = Cs^{2-2a}M_0(y),$$

where M_0 is M on the base link. This majorant is integrable with respect to $d\tau\, dV_{D_0}$. The base factor M_0 is integrable by (2.11). The factor s^{2-2a} is integrable since s vanishes linearly at both ends and $2-2a = 21/25 > -1$.

We can now pass to the limit in (2.10) in both dimensions. At each point where $w > 0$, the cutoff is eventually one with zero gradient. The error is already zero where $w = 0$. The preceding integrable majorants therefore give $\int_D \mathcal{E}_{\xi_j} \to 0$. On the left, $0 \le \xi_j^2 w^{2a} \le w^{2a}$, and w^{2a} is integrable on the smooth link by boundedness. On the suspension its density in the product coordinates is $s^{4-2a} w_0^{2a}$ with $4-2a = 71/25 > -1$, so it too is integrable. Dominated convergence gives $c\int_D w^{2a}$ as the limit on the left. This is positive because a nonflat component has $w > 0$ on an open set. The two limits contradict (2.10).

Thus $T = 0$ on each positive link component D in dimensions five and six. The gradient formula and the Hessian calculation show that $\theta \mapsto Du(\theta)$ has zero derivative in every tangent direction in D, so it is a constant vector e there. Taking its scalar product with θ gives $g(\theta) = e \cdot \theta$, and (2.1) gives $|e| = 1$ on taking the boundary trace of Du. By continuity $g = 0$ on ∂D, so D has no boundary in the open hemisphere $\{e \cdot \theta > 0\}$. Since that hemisphere is connected, D fills it. Any two open hemispheres with normals e, f intersect at $(e+f)/|e+f|$ if $e + f \neq 0$. Opposite hemispheres would put positive phase on both sides of the same equator, contrary to classicality at every point of the equator. Hence there is one positive hemisphere and $u = (e \cdot x)_+$. With the known exclusions through dimension four this proves $n_* > 6$.

3. A scalar defect for the gradient

Throughout this section and the next, the ambient dimension is six. In the positive phase Ω, put $p = Du$, $B = D^2 u$, $w = |B|$, and $\sigma = 1 - |p|^2$. For the local estimate we assume $|p| \le 1$ in the positive phase of the ball under study, so $0 \le \sigma \le 1$ there. For a global classical solution the gradient bound follows from [4, Lemma 3.2].

Harmonicity gives $\Delta p = D\Delta u = 0$ and $\Delta |p|^2 = 2p \cdot \Delta p + 2|Dp|^2 = 2w^2$. On Γ, use the boundary Hessian identity $BN = -HN$ from Appendix A. Since $p = N$ and $\nu = -N$, it gives $B\nu = Hp$. In particular, $\partial_\nu \sigma = -2p \cdot B\nu = -2H$. The bulk and boundary identities are

$$\Delta p = 0, \qquad \Delta \sigma = -2w^2, \qquad \partial_\nu p = Hp, \qquad \partial_\nu \sigma = -2H, \tag{3.1}$$

where the last two hold on Γ. The components of p are the Jacobi fields from translations. Also, for $x \in \Gamma$ the point $x - t\nu(x)$ lies in Ω for small $t > 0$. Since $\sigma(x) = 0$ and $\sigma(x - t\nu(x)) \ge 0$, the outward derivative satisfies

$$-2H(x) = \partial_\nu \sigma(x) = \lim_{t \downarrow 0} \frac{\sigma(x) - \sigma(x - t\nu(x))}{t} \le 0.$$

Thus $H \ge 0$ on the free boundary in the ball.

For the fixed unit direction e, let P_e project onto $e^\perp$. Suppressing the dependence on e, write the defect from Section 1.2 as the length of $F = (P_e p, \sigma/\sqrt{30})$. Set

$$h^2 = |F|^2 = |P_e p|^2 + \frac{\sigma^2}{30}, \qquad h_\delta = (h^2 + \delta^2)^{1/2}, \qquad 0 < \delta \le 1. \tag{3.2}$$

Figure 3 shows both contributions to the defect. Regularizing the length makes h_δ smooth at $F = 0$. The derivative of a length sees only the projection of each derivative onto the vector's direction. The trace-free condition on B lets the complementary terms control the full Hessian.

Lemma 3.1. *The regularized defect satisfies*

$$h_\delta \Delta h_\delta \ge \frac{w^2}{75}, \qquad |Dh_\delta|^2 \le \frac{17}{15} w^2 \quad \text{in } \Omega, \qquad \partial_\nu h_\delta = H\left(h_\delta - \frac{\delta^2}{h_\delta}\right) \quad \text{on } \Gamma.$$

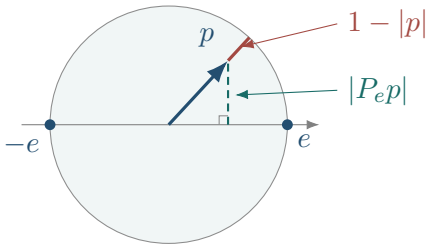


FIGURE 3. A planar slice of the unit ball of gradient values. The dashed segment measures $|P_e p|$. The radial gap measures $1-|p|$, with $\sigma = (1-|p|)(1+|p|)$.

Proof. Since P_e is constant, $D(P_e p) = P_e B$ and $\Delta(P_e p) = 0$. Also $D\sigma = -2Bp$ by symmetry of B. Thus

$$|DF|^2 = |P_e B|^2 + \frac{2}{15}|Bp|^2, \qquad F \cdot \Delta F = \frac{\sigma \Delta \sigma}{30} = -\frac{\sigma}{15} w^2.$$

Differentiating $h_\delta^2 = |F|^2 + \delta^2$ once and twice yields

$$h_\delta D h_\delta = (DF)^{\mathsf{T}} F = B(P_e p) - \frac{\sigma}{15} Bp,$$
$$h_\delta \Delta h_\delta = |DF|^2 - |Dh_\delta|^2 - \frac{\sigma}{15} w^2.$$

Since $|F| \le h_\delta$, Cauchy–Schwarz in the first identity gives $|Dh_\delta|^2 \le |DF|^2$. Both $|P_e B| \le w$ and $|Bp| \le w|p| \le w$, so $|Dh_\delta|^2 \le (17/15)w^2$.

If $h = 0$, then $p = \pm e$, $\sigma = 0$, and $Dh_\delta = 0$. In that case the Laplacian identity gives

$$h_\delta \Delta h_\delta = |P_e B|^2 + \tfrac{2}{15}|Be|^2 \ge \tfrac{2}{15} w^2,$$

because $|P_e B|^2 + |Be|^2 = w^2$ by orthogonal projection and symmetry of B.

At a point with $h > 0$, put $s = |P_e p|$ and choose an orthonormal basis $a, \xi_1, \dots, \xi_4$ of $e^\perp$ with $P_e p = sa$. If $s = 0$, any such basis works. Symmetry gives $|P_e B|^2 = |Ba|^2 + \sum_{j=1}^4 |B\xi_j|^2$. Set $U = Ba$, $V = 2Bp/\sqrt{30}$, $\alpha = s$, and $\beta = \sigma/\sqrt{30}$. Then $Dh_\delta = (\alpha U - \beta V)/h_\delta$ and $h_\delta^2 \ge h^2 = \alpha^2 + \beta^2$. Expanding the squares gives

$$(\alpha^2 + \beta^2)(|U|^2 + |V|^2) - |\alpha U - \beta V|^2 = |\beta U + \alpha V|^2.$$

It follows that

$$|DF|^2 - |Dh_\delta|^2 \ge \sum_{j=1}^4 |B\xi_j|^2 + \frac{|\sigma Ba + 2sBp|^2}{30h^2}.$$

Because $a \cdot p = s$, the squared norm of $\sigma a + 2sp$ is $\sigma^2 + 4\sigma s^2 + 4s^2(1-\sigma) = \sigma^2 + 4s^2 > 0$. Normalize this vector to $v = (\sigma a + 2sp)/|\sigma a + 2sp|$. It lies in $\operatorname{span}\{a, e\}$, so it is orthogonal to the four ξ_j. We have proved

$$h_\delta \Delta h_\delta \ge E - \frac{\sigma}{15} w^2, \qquad E = \lambda |Bv|^2 + \sum_{j=1}^4 |B\xi_j|^2, \qquad \lambda = \frac{\sigma^2 + 4s^2}{\sigma^2 + 30 s^2} \in (0, 1].$$

Complete these five directions to an orthonormal basis with v at index 0 and the last direction at index 5. If $b_i = B_{ii}$, the diagonal and off-diagonal parts of E in this basis are

$$E_{\mathrm{diag}} = \lambda b_0^2 + \sum_{i=1}^4 b_i^2, \qquad E_{\mathrm{off}} = \sum_{0 \le i < j \le 5} (\omega_i + \omega_j) B_{ij}^2, \qquad \omega = (\lambda, 1, 1, 1, 1, 0).$$

The weights add because $B_{ij} = B_{ji}$. Each pair has at least one index in a measured column, so $2\sum_{i<j} B_{ij}^2 \le (2/\lambda)E_{\mathrm{off}}$. The omitted diagonal entry is controlled by $\operatorname{tr} B = 0$. Indeed, Cauchy–Schwarz with weights ω_i for $i \le 4$ gives

$$b_5^2 = \left(\sum_{i=0}^{4} b_i\right)^2 \le (1/\lambda + 4)E_{\mathrm{diag}}, \qquad \sum_{i=0}^{4} b_i^2 \le E_{\mathrm{diag}}/\lambda.$$

Adding the diagonal and off-diagonal bounds yields $w^2 \le (2/\lambda + 4)E$.

To absorb the term $-\sigma w^2/15$, use $s^2 \le 1 - \sigma$. As a function of $r = s^2$, the fraction

$$\frac{\lambda}{2 + 4\lambda} = \frac{\sigma^2 + 4r}{6\sigma^2 + 76r}$$

has derivative $-52\sigma^2/(6\sigma^2+76r)^2$ wherever the denominator is positive. Thus it is bounded below by its value at $r = 1 - \sigma$. To compare this value minus $\sigma/15$ with $1/75$, multiply the difference by the positive factor $75(6\sigma^2 + 76(1 - \sigma))$. The result is

$$\begin{aligned} &75(\sigma^2 + 4(1 - \sigma)) - (5\sigma + 1)(6\sigma^2 + 76(1 - \sigma)) \\ &\quad = 224 - 604\sigma + 449\sigma^2 - 30\sigma^3 \\ &\quad \ge 224 - 604\sigma + 419\sigma^2 = 419\left(\sigma - \frac{302}{419}\right)^2 + \frac{2652}{419} > 0. \end{aligned}$$

The inequality uses $0 \le \sigma \le 1$. Consequently $E - \sigma w^2/15 \ge w^2/75$, as required.

On Γ, $\sigma = 0$, $\partial_\nu(P_e p) = HP_e p$, and $H \ge 0$. Differentiating h_δ^2 in direction ν and dividing by $2h_\delta > 0$ gives

$$\partial_\nu h_\delta = \frac{F \cdot \partial_\nu F}{h_\delta} = \frac{H|P_e p|^2}{h_\delta} = H\left(h_\delta - \frac{\delta^2}{h_\delta}\right) \le H h_\delta. \qquad \square$$

4. Stability and an iteration

The iteration uses the stability inequality to estimate powers of the defect on smaller balls. We work under the bound $|p| = |Du| \le 1$ from Section 3 in the positive phase of the ambient ball. The free-boundary term has the right sign for a function satisfying a Robin inequality. We first derive the energy and Sobolev bounds, then construct powers that preserve this sign.

4.1. **Energy and Sobolev inequalities.** Let $G \ge 0$ be locally $C^{1,1}$ up to the free boundary and satisfy $\partial_\nu G \le HG$ on Γ. Let η be smooth and compactly supported in the ambient ball. Near the support of η, flatten the free boundary in finitely many charts and reflect ηG across the flattened boundary. A partition of unity combines these extensions with the interior pieces to give a Lipschitz ambient extension compactly supported in the ball. Mollification approximates it uniformly and in H^1. On the compact boundary portion meeting these supports, H is bounded and the surface measure is finite. Hence the boundary integrals also converge, so ηG is an admissible test in (1.2).

The identity $|D(\eta G)|^2 = G^2|D\eta|^2 + DG \cdot D(\eta^2 G)$ follows by expanding both products. Integrate the divergence of the locally Lipschitz field $\eta^2 G DG$. Its boundary flux is $\eta^2 G \partial_\nu G$, and compact support leaves only the free boundary in the flux integral. Thus

$$0 \le Q(\eta G) = \int_\Omega G^2|D\eta|^2 - \int_\Omega \eta^2 G \Delta G + \int_\Gamma \eta^2 G(\partial_\nu G - HG).$$

The last term is nonpositive. Here ΔG is the weak Laplacian, and $\partial_\nu G$ is taken from the positive side. We obtain

$$\int_\Omega \eta^2 G \Delta G \le \int_\Omega G^2|D\eta|^2. \tag{4.1}$$

For the Sobolev estimate, take f locally Lipschitz up to the free boundary with compact support in the ambient ball. Put $q = |f|^{5/2}$ on Ω and extend it by zero to $\widetilde{q}$ on $\mathbb{R}^6$. The Bernoulli condition bounds the jump of this extension. Indeed, $\operatorname{div}(qDu) = Du \cdot Dq$ in Ω, and its outward flux on Γ is $q\partial_\nu u = -q$. Therefore

$$\int_\Gamma q = -\int_\Omega Du \cdot Dq \le \int_\Omega |Dq|.$$

For a smooth compactly supported vector field X, integrating $\operatorname{div}(qX) = Dq \cdot X + q \operatorname{div} X$ gives

$$-\int_{\mathbb{R}^6} \widetilde{q} \operatorname{div} X = \int_\Omega Dq \cdot X - \int_\Gamma q\, X \cdot \nu.$$

Thus $D\widetilde{q}$ has an interior part $Dq\,dx$ and a boundary part $-q\nu\,dS$, with total variation bounded by $\int_\Omega |Dq| + \int_\Gamma q \le 2\int_\Omega |Dq|$.

For a nonnegative smooth mollifier ρ_ϵ of integral one, the derivative of $\widetilde{q} * \rho_\epsilon$ is $D\widetilde{q} * \rho_\epsilon$. Its L^1 norm is bounded by the total variation of $D\widetilde{q}$, by integration against ρ_ϵ. Apply the Euclidean $W^{1,1}$ Sobolev inequality to the mollifications and use Fatou's lemma as they converge almost everywhere to $\widetilde{q}$. Since $(5/2)(6/5) = 3$ and $|Dq| \le (5/2)|f|^{3/2}|Df|$, this gives

$$\|f\|^{5/2}_{L^3(\Omega)} = \|\widetilde{q}\|_{L^{6/5}(\mathbb{R}^6)} \le C\int_\Omega |f|^{3/2}|Df| \le C\|f\|^{3/2}_{L^3(\Omega)}\|Df\|_{L^2(\Omega)}.$$

By Cauchy–Schwarz we obtain the last inequality. Dividing when $\|f\|_{L^3(\Omega)} > 0$ proves the estimate

$$\|f\|_{L^3(\Omega)} \le C\|Df\|_{L^2(\Omega)}. \tag{4.2}$$

In particular, if $|DG|^2 \le AG\Delta G$ almost everywhere for some $A > 0$, then (4.1) gives

$$\int_\Omega |D(\eta G)|^2 \le 2\int_\Omega (\eta^2|DG|^2 + G^2|D\eta|^2) \le 2(A+1)\int_\Omega G^2|D\eta|^2.$$

Applying (4.2) to ηG gives

$$\|\eta G\|^2_{L^3(\Omega)} \le C(A+1)\int_\Omega G^2|D\eta|^2. \tag{4.3}$$

4.2. **Raising the power of the defect.** Near a point of Γ where $h > 0$, $h_\delta \to h$ in C^1. The Robin identity in Lemma 3.1 therefore gives $\partial_\nu h = Hh$, so $\partial_\nu(h^k) - Hh^k = (k-1)Hh^k$. For $k > 1$ this can be positive. To preserve the boundary inequality when taking powers, divide by a positive multiplier with the same Robin coefficient. The boundary identities $\sigma = 0$ and $\partial_\nu\sigma = -2H$ suggest $b = 1 - \sigma/2$, so set

$$b = \frac{1+|p|^2}{2}, \qquad t = \frac{h_\delta}{b}, \qquad G = bF(t). \tag{4.4}$$

We have $1/2 \le b \le 1$, $Db = Bp$, and hence $|Db| \le w$. Equation (3.1) gives $\Delta b = w^2$. On Γ, $b = 1$ and $\partial_\nu b = H = Hb$. The quotient rule and Lemma 3.1 give there

$$\partial_\nu t = \frac{\partial_\nu h_\delta - t\partial_\nu b}{b} = -H\delta^2/h_\delta.$$

For nonnegative C^1 functions F with $F' \ge 0$, the boundary inequality for $G = bF(t)$ follows from

$$\partial_\nu G - HG = bF'(t)\partial_\nu t = -HF'(t)\delta^2/h_\delta \le 0. \tag{4.5}$$

Here we used $H \ge 0$ from Section 3.

For a twice differentiable F, the bulk product rule gives

$$\Delta G = F(t)\Delta b + F'(t)(b\Delta t + 2Db \cdot Dt) + bF''(t)|Dt|^2.$$

Since $bt = h_\delta$, the middle bracket equals $\Delta h_\delta - t\Delta b$. With $\Delta b = w^2$, this becomes

$$\Delta G = F'(t)\Delta h_\delta - (tF'(t) - F(t))w^2 + bF''(t)|Dt|^2. \tag{4.6}$$

The term $-(tF' - F)w^2$ must be absorbed by $F'\Delta h_\delta$. For a power $F(t) = t^k$, the ratio $(tF' - F)/F'$ is $(1 - 1/k)t$, so its size grows with t. Use the power up to $d = (150\sqrt{2})^{-1}$ and continue by its tangent line. For $k = 1$ or $k \geq 3/2$, define

$$F_k(t) = \begin{cases} t^k, & 0 \leq t \leq d, \\ d^k + kd^{k-1}(t - d), & t > d. \end{cases}$$

The continuation keeps F_k convex and holds the ratio at its value at d for $t \geq d$. For $t > 0$, set $T_d = \min(t, d)$. Figure 4 illustrates the continuation and the clipped power for $k = 2$.

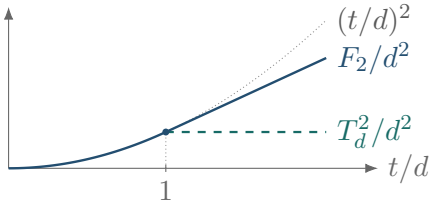


FIGURE 4. The power, its tangent-line continuation, and the clipped power for $k = 2$. All three coincide for $t \leq d$. The continuation preserves the slope at d, while the clipped power stays constant above d.

In terms of T_d, the two branches give

$$F_k(t) = T_d^k + kT_d^{k-1}(t - T_d), \qquad F_k'(t) = kT_d^{k-1},$$
$$tF_k'(t) - F_k(t) = (k - 1)T_d^k = (1 - 1/k)T_d F_k'(t).$$

In particular, $0 \leq tF_k' - F_k \leq dF_k'$. Since $h^2 \leq 1 - \sigma + \sigma^2/30 \leq 1$ and $0 < \delta \leq 1$, we have $h_\delta \leq \sqrt{2}$ and $\delta \leq t \leq 2\sqrt{2}$. Thus F_k' is Lipschitz on the range of t for fixed $\delta > 0$, so $G = bF_k(t)$ is locally $C^{1,1}$ up to Γ.

The product rule (4.6) holds almost everywhere for this G. At $t = d$ we may set $F_k''(d) = 0$. The points on that level set with $Dt \neq 0$ lie locally on smooth hypersurfaces and form a null set. At a point x of the level with $Dt(x) = 0$, $t(y) - d = o(|y - x|)$. The Lipschitz bound for F_k' then gives $F_k'(t(y)) - F_k'(d) = o(|y - x|)$, so the derivative of $F_k'(t)$ at x is zero. This justifies the chain rule on the rest of the level. Local $C^{1,1}$ regularity ensures that the weak Laplacian is represented by this almost-everywhere formula.

By Lemma 3.1, $\Delta h_\delta \geq w^2/(75\sqrt{2}) = 2dw^2$. Since $tF_k' - F_k \leq dF_k'$, (4.6) now gives

$$\Delta G \geq dF_k' w^2 + bF_k''|Dt|^2. \tag{4.7}$$

We compare $|DG|^2$ with $G\Delta G$. For $k = 1$, $F_1(t) = t$ and $G = h_\delta$. The two bulk bounds in Lemma 3.1 give $|DG|^2 \leq 85G\Delta G$, since $75(17/15) = 85$.

For $k \geq 3/2$ and $t < d$, differentiate $G = bF_k(t)$ and use $(F_k')^2/(F_k F_k'') = k/(k - 1) \leq 3$ and $F_k/F_k' = t/k \leq d$. Thus

$$|DG|^2 \leq 2b^2(F_k')^2|Dt|^2 + 2F_k^2 w^2 \leq 6b^2 F_k F_k''|Dt|^2 + 4d\, bF_k F_k' w^2 \leq 6G\Delta G.$$

The middle inequality uses $b \geq 1/2$, and the last uses (4.7). For $t \geq d$, differentiate $bt = h_\delta$ to get

$$DG = F_k' Dh_\delta - (tF_k' - F_k)Db.$$

The bounds on Dh_δ and Db yield $|DG| \leq (\sqrt{17/15} + d)F_k' w$. Also $F_k' \leq (k/d)F_k$ there, since $F_k' = kd^{k-1}$ and $F_k \geq d^k$. As $G\Delta G \geq (d/2)F_k F_k' w^2$, we obtain $|DG|^2 \leq 2(\sqrt{17/15} + d)^2 kG\Delta G/d^2$. Combining both branches and the case $k = 1$ gives

$$|DG|^2 \leq C_d k\, G\Delta G \tag{4.8}$$

almost everywhere.

The formula for F_k gives $F_k/T_d^k = 1 + k(t - T_d)/T_d$. The second term is zero for $t \le d$. For $t > d$, $T_d = d$ and $t \le 2\sqrt{2}$, so $1 \le F_k/T_d^k \le 1 + 2\sqrt{2}k/d$ everywhere. With $1/2 \le b \le 1$, we have $\frac{1}{2}T_d^k \le G \le C_d k T_d^k$. When the ambient ball is B_1, choose a cutoff $0 \le \eta \le 1$ that is one on B_r, supported in B_ρ, and satisfies $|D\eta| \le C/(\rho - r)$, where $0 < r < \rho \le 1$. Using (4.3) and (4.8),

$$\begin{aligned}\|T_d\|_{L^{3k}(\Omega\cap B_r)}^{2k} &\le \|\eta T_d^k\|_{L^3(\Omega)}^2 \le 4\|\eta G\|_{L^3(\Omega)}^2 \\ &\le C_d k \int_\Omega G^2 |D\eta|^2 \le \frac{C_d k^3}{(\rho - r)^2}\|T_d\|_{L^{2k}(\Omega\cap B_\rho)}^{2k}.\end{aligned}$$

Taking $2k$-th roots gives the common iteration step

$$\|T_d\|_{L^{3k}(\Omega\cap B_r)} \le \left(\frac{C_d k^3}{(\rho - r)^2}\right)^{1/(2k)} \|T_d\|_{L^{2k}(\Omega\cap B_\rho)} \qquad (k = 1 \text{ or } k \ge 3/2). \tag{4.9}$$

Proof of Theorem 1.2. Set $y = x_0 + Rx$ and $u_R(x) = u(y)/R$ on B_1. Then $D_x u_R(x) = D_y u(y)$ and $\Delta_x u_R(x) = R\Delta_y u(y)$, so the gradient bound and the Bernoulli problem are preserved. For a test ϕ in B_1, let $\Phi(y) = \phi((y - x_0)/R)$. In the bulk term, $D_x\phi = RD_y\Phi$ and $dx = R^{-6}dy$. In the boundary term, $H_R(x) = RH(y)$ and $dS_x = R^{-5}dS_y$, since the inward normals agree and lengths scale by R^{-1}. Thus $Q_{u_R}(\phi) = R^{-4}Q_u(\Phi) \ge 0$. We now use Ω and h for the rescaled positive phase and defect. The gradient is unchanged at corresponding points, so $\|h\|_{L^2(\Omega\cap B_1)} = E_e$ by the same volume change.

Take $k_j = (3/2)^j$ and $r_j = 1/2 + 2^{-j-1}$ for $j \ge 0$. Then $k_0 = 1$, $r_0 = 1$, and $3k_j = 2k_{j+1}$. Apply (4.9) from r_j to r_{j+1}. The gap is 2^{-j-2}, so the logarithm of the factor at step j is

$$\frac{\log C_d + 3j\log(3/2) + (2j+4)\log 2}{2k_j}.$$

The sum of these logarithms is finite because $\sum_{j\ge0}(j+1)(2/3)^j < \infty$. Hence the products of the iteration factors are bounded by a dimensional constant. Since $T_d \le t \le 2h_\delta$, we obtain for all j

$$\|T_d\|_{L^{2k_j}(\Omega\cap B_{1/2})} \le C\|h_\delta\|_{L^2(\Omega\cap B_1)}.$$

Write M_δ for the right-hand side. If $T_d > M_\delta$ at a point of $\Omega \cap B_{1/2}$, continuity gives a ball U about that point in this set and a number $a > M_\delta$ with $T_d \ge a$ on U. The last norm is then at least $a|U|^{1/(2k_j)}$, which tends to $a > M_\delta$ and contradicts the norm bound. Thus $T_d \le M_\delta$ pointwise.

Letting $\delta \downarrow 0$ yields $\min(h/b, d) \le CE_e$, because $\|h_\delta\|_{L^2(\Omega\cap B_1)}^2 = E_e^2 + \delta^2|\Omega \cap B_1|$. Choose $\varepsilon_0 = d/(2C)$. If $E_e \le \varepsilon_0$, then $CE_e \le d/2 < d$, so the minimum equals h/b throughout $\Omega \cap B_{1/2}$. As $b \le 1$, this bounds $h \le CE_e$ there. The gradient is unchanged by the rescaling and $B_{1/2}$ corresponds to $B_{R/2}(x_0)$, which proves the stated estimate. □

5. Blow-down and classification

The direction supplied by the blow-down may vary with the radius. The length defect $1 - |Du|^2$ is independent of that direction, so it can be estimated at a fixed point as the radius grows.

Proof of Theorem 1.1. If $\Gamma = \varnothing$, then Ω is both open and closed in $\mathbb{R}^6$. By connectedness, either $\Omega = \varnothing$ and $u = 0$, or $\Omega = \mathbb{R}^6$ and u is an entire nonnegative harmonic function, hence constant by Liouville's theorem.

Assume $\Gamma \ne \varnothing$ and translate a free-boundary point to the origin. Suppose that $D^2u(x_*) \ne 0$ at some $x_* \in \Omega$. For $L > 0$, set $u_L(x) = u(Lx)/L$. In its positive phase,

$$Du_L(x) = Du(Lx), \qquad D^2u_L(x) = LD^2u(Lx).$$

In particular, harmonicity and the unit boundary gradient are preserved. The scaling of the stability form in the proof of Theorem 1.2 gives

$$Q_{u_L}(\phi) = L^{-4} Q_u(\Phi), \qquad \Phi(y) = \phi(y/L), \qquad \phi \in C_c^\infty(\mathbb{R}^6).$$

The dilated test Φ is compactly supported, so the right-hand side is nonnegative. Appendix A then gives inner stability of u_L. Choose $L > \max\{|x_*|, |D^2u(x_*)|^{-1}\}$. At x_*/L, which is inside the positive phase of u_L and in B_1, the rescaled Hessian has norm $L|D^2u(x_*)| > 1$. By continuity it has norm greater than one on a neighborhood of that point. In the blow-down argument write u, Ω, Γ for the rescaled solution, positive phase, and free boundary.

We have $0 \in \Gamma$, $\|D^2u\|_{L^\infty(\Omega \cap B_1)} \geq 1$, and $6 < n_*$ by Section 2. Thus [9, Proposition 4.12] supplies, for each $R > 0$, a unit vector e_R and a profile $V_R(x) = |e_R \cdot x|$ such that

$$R^{-6} \int_{B_R} |Du - DV_R|^2 \leq \omega(R^{-1})^2 \longrightarrow 0 \qquad \text{as } R \to \infty. \tag{5.1}$$

Here ω is the modulus in that proposition and the gradients are weak gradients on the whole ball. The global gradient bound gives $|Du| \leq 1$ [4, Lemma 3.2]. Almost everywhere DV_R equals e_R or $-e_R$, so the comparison in Section 1.4 applies with $q = DV_R$. The normalized defect on B_R therefore satisfies

$$E_R^2 := R^{-6} \int_{\Omega \cap B_R} h_{e_R}^2 \leq (1 + 4/30) R^{-6} \int_{B_R} |Du - DV_R|^2.$$

By (5.1), $E_R \to 0$, so $E_R \leq \varepsilon_0$ for all sufficiently large R. Theorem 1.2 then gives

$$\sup_{\Omega \cap B_{R/2}} h_{e_R} \leq C E_R.$$

Fix $x \in \Omega$. As soon as $R > 2|x|$ and the estimate applies,

$$0 \leq \sigma(x) = 1 - |Du(x)|^2 \leq \sqrt{30}\, h_{e_R}(x) \leq \sqrt{30}\, C E_R.$$

The upper bound tends to zero and $\sigma(x)$ is independent of R. Hence $\sigma = 0$ on Ω. By (3.1), $0 = \Delta\sigma = -2|D^2u|^2$ there, contradicting the choice of the rescaled point. Thus the Hessian of the original solution vanishes throughout its positive phase.

Return to the original solution u, and let U be a connected component of its positive phase. The equation $D^2u = 0$ makes Du locally constant, hence constant on U. It follows that $u = \ell_U$ there for an affine function $\ell_U(x) = e_U \cdot x - c_U$. Since the free boundary is nonempty, U is a proper open subset of $\mathbb{R}^6$, so ∂U is nonempty. If $z \in \partial U$ had $u(z) > 0$, a small connected ball about z in Ω would meet U and lie in U, contradicting $z \in \partial U$. Therefore $\partial U \subset \Gamma$. At a point of ∂U, the positive-side trace of Du equals e_U, so the Bernoulli condition gives $|e_U| = 1$. Continuity gives $\ell_U = 0$ on ∂U. Thus U lies in the half-space $\{\ell_U > 0\}$ and has no relative boundary there. It fills that half-space by connectedness.

If two of these half-spaces have normals e_1, e_2 with $e_1 + e_2 \neq 0$, then $e_i \cdot (e_1 + e_2) = 1 + e_1 \cdot e_2 > 0$ for $i = 1, 2$. Consequently large positive multiples of $e_1 + e_2$ belong to both half-spaces. Distinct components are disjoint, so their normals must be opposite. There can be at most two components, since two components with normals opposite to a given one would have the same normal and intersect. A single component gives $u = (e_U \cdot x - c_U)_+$. With two, write the components as $\{e \cdot x > b\}$ and $\{e \cdot x < a\}$. A point with $b < e \cdot x < a$ would lie in both, so disjointness gives $a \leq b$. If $a = b$, the positive phase occupies both sides locally at each point of the common boundary plane, contrary to classicality. Thus $a < b$. The affine values on the two components and the zero values on the intervening strip give the second profile in Theorem 1.1. Figure 1 shows both profiles. □

Proof of Corollary 1.3. For $1 \leq n \leq 5$, set $U(x, y) = u(x)$ for $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^{6-n}$. Its positive phase and free boundary are $\Omega \times \mathbb{R}^{6-n}$ and $\Gamma \times \mathbb{R}^{6-n}$. The product boundary is smooth with

a single positive side locally. The identities $\Delta U = \Delta u$ and $DU = (Du, 0)$ give harmonicity and preserve the boundary gradient length, so U is classical. Its inward unit normal is $(N(x), 0)$. The tangential divergence in the original directions is $H(x)$ and the derivatives in the added directions vanish. Thus the product boundary has mean curvature $H(x)$.

For $n \geq 2$, scalar stability of u follows from (1.2). For $n = 1$, the boundary has zero-dimensional tangent space, so $H = 0$ and the scalar form is $Q_u(\psi) = \int_\Omega |\psi'|^2 \geq 0$. For $\phi \in C_c^\infty(\mathbb{R}^6)$, each slice $\phi_y = \phi(\cdot, y)$ has compact support in the x-projection of $\operatorname{supp}\phi$. The product calculation of Section 2.2, with the surface measure $dS_x\, dy$ and the Dirichlet integrand $|D_x\phi|^2 + |D_y\phi|^2$, gives

$$Q_U(\phi) = \int_{\mathbb{R}^{6-n}} Q_u(\phi_y)\, dy + \int_{\Omega\times\mathbb{R}^{6-n}} |D_y\phi|^2 \geq 0.$$

Appendix A gives inner stability of U.

Apply Theorem 1.1. Its Hessian conclusion gives $D^2u(x) = D_x^2U(x, 0) = 0$ in Ω. The free boundary of U is empty exactly when Γ is empty, and in this case the constant conclusion passes to u. Otherwise one of the two nonconstant profiles holds for U. On each positive component its gradient is e or $-e$. Since $D_yU = 0$, the unit vector e has the form $(e_0, 0)$ with $e_0 \in \mathbb{R}^n$ a unit vector. Restricting the profile to $y = 0$ gives the claimed formula for u. □

6. Local curvature and finite index

6.1. **Bounds on balls centered at the free boundary.** A ball containing a free-boundary point has a local bound on the gradient. The Bernoulli condition controls the solution near that point, and Harnack chains carry this control into the positive phase. Precisely, if v is classical in $B_2 \subset \mathbb{R}^n$, $n \geq 2$, and $0 \in \Gamma_v$, then

$$\sup_{B_{3/2}} v + \sup_{B_{3/2}\cap\{v>0\}} |Dv| \leq L_n. \tag{6.1}$$

Take $x \in B_{3/2} \cap \{v > 0\}$ with $d = \operatorname{dist}(x, \Gamma_v) \leq 1/16$. A nearest point $p \in \Gamma_v$ exists, since the infimum can be taken in $\overline{B_{1/8}(x)} \subset B_2$. The connected ball $B_d(x)$ contains x and avoids Γ_v, so it is positive. Harnack's inequality gives $v \geq c_n v(x)$ on $\partial B_{d/2}(x)$. On the annulus, let $w(s)$ be the radial harmonic function with values $c_n v(x)$ at $s = d/2$ and zero at $s = d$, where s is distance from x. Thus

$$w(s) = \begin{cases} c_n v(x) \dfrac{s^{2-n} - d^{2-n}}{(d/2)^{2-n} - d^{2-n}}, & n > 2, \\ c_2 v(x) \dfrac{\log(d/s)}{\log 2}, & n = 2. \end{cases}$$

The boundary values give $v \geq w$ in the annulus by the maximum principle. Minimizing distance on the smooth free boundary makes $x - p$ orthogonal to its tangent space. Since the segment from p toward x is positive, $N(p) = (x - p)/d$. Along this segment $v \geq w$ and both vanish at p, so $1 = \partial_N v(p) \geq -w'(d)$. Differentiation gives

$$-w'(d) = \begin{cases} \dfrac{c_n(n-2)}{2^{n-2} - 1} \dfrac{v(x)}{d}, & n > 2, \\ \dfrac{c_2}{\log 2} \dfrac{v(x)}{d}, & n = 2. \end{cases}$$

Thus $v(x) \leq C_n d$. The interior gradient estimate and Harnack's inequality give

$$|Dv(x)| \leq \frac{C_n}{d} \sup_{B_{d/2}(x)} v \leq \frac{C_n}{d} v(x) \leq C_n.$$

If $d > 1/16$, the distance to Γ_v along the segment from x to 0 is continuous and ends at zero. Stop at the first point with distance $1/16$. The segment up to there stays in the positive phase,

since an exit would require distance zero. Divide the segment into at most 96 equal steps of length at most $1/64$, since its length is at most $3/2$. Use the endpoints of these steps as centers. The balls of radius $1/32$ about them are positive and contained in B_2. Each next center is in the inner half of the preceding ball, so Harnack's inequality transfers the bound at the last center to x. Applying the interior gradient estimate and Harnack's inequality on $B_{1/32}(x)$ bounds $|Dv(x)|$. Together with $v = 0$ on the zero phase this proves (6.1).

The rescaling argument needs a Hessian bound on neighborhoods that grow to fill the whole space at the new scale. In taking the limit we must allow two free-boundary sheets to touch.

Let v_j be classical stable in B_{S_j}, with $S_j \to \infty$, $0 \in \Gamma_{v_j}$, and $|D^2 v_j| \le M$ in the positive phase. Pass to a subsequence with $S_j \ge j$ and restrict to B_j to apply [9, Lemma 4.6]. The local Lipschitz hypothesis follows from classicality as in Section 1. The lemma gives a global 1-Lipschitz, inner stable limit v, with locally uniform and strong H^1_{loc} convergence. Both the closed positive phases and the free boundaries converge locally in Hausdorff distance. In particular $0 \in \Gamma_v$. On each ball compactly contained in $\{v > 0\}$ the v_j are eventually positive by locally uniform convergence. Interior harmonic estimates there give harmonicity of v and convergence of all derivatives on smaller balls.

The Hessian bound controls how quickly the gradient can turn along a segment in the positive phase. Fix a prelimit boundary point p in a bounded set and take j large enough that the rays below stay in B_{S_j}. If b is a unit vector with $b \cdot N(p) \ge 1/16$, the ray in direction b enters the positive phase. Its initial value is zero and its initial gradient is $N(p)$. Integrating the Hessian bound up to a possible first exit gives

$$v_j(p + sb) \ge s\, b \cdot N(p) - Ms^2/2, \qquad |Dv_j(p + sb) - N(p)| \le Ms.$$

For $0 < s \le \ell = 1/(16(1 + M))$ the lower bound is at least $s/32$. The inequalities hold up to a first exit by continuity, and the strictly positive value there rules out such an exit.

Take another boundary point q, with $0 < d = |p - q| \le r_0 = 1/(10^4(1 + M))$. If $N(p) \cdot N(q) \ge -31/32$, set $b = (N(p) + N(q))/|N(p) + N(q)|$. Both scalar products with b equal $\sqrt{(1 + N(p) \cdot N(q))/2} \ge 1/8$. The point $p + 32db$ is within $33d < \ell$ of both p and q. Its displacement from q has scalar product at least $32d/8 - d = 3d$ with $N(q)$, so the corresponding unit direction has scalar product at least $3/33 > 1/16$. Both rays thus reach the same point through the positive phase. Comparing the two initial gradients with the gradient at this point gives $|N(p) - N(q)| \le M(32d + 33d) = 65Md$. Since $d \le r_0$,

$$N(p) \cdot N(q) = 1 - \tfrac{1}{2}|N(p) - N(q)|^2 \ge 1 - \tfrac{1}{2}(65/10^4)^2 > 31/32.$$

Including the case of nearly opposite normals, we obtain

$$|N(p) \cdot N(q)| \ge 31/32.$$

Use $N(p)$ as the vertical direction and set $r = r_0/16$. In the cylinder $|x'| < 4r$, $|x_n| < 4r$ about p, each boundary point has distance less than r_0 from p. Its normal has $|N_n| \ge 31/32$, so it lies on a local graph of slope $|N'|/|N_n| \le \sqrt{63}/31 < 1/3$. There is at most one point of each orientation on any vertical line in this cylinder. Indeed, if two inward normals point up, the inward vertical ray from the lower point reaches the upper point within length ℓ while still in the positive phase. For two downward normals use the ray from the upper point. The same reasoning shows that when both orientations occur, the upward-facing graph lies above the downward-facing graph.

Each sheet meeting $|x'| < r$, $|x_n| < r$ extends as a graph over $|x'| < 2r$. Continue its local graph along a straight segment in the base. The length at most $3r$ and slope less than $1/3$ keep the height within $2r$. A stopping point would therefore lie inside the larger cylinder, where closedness of the free boundary and $|N_n| \ge 31/32$ allow the implicit function theorem to continue the graph. The sign of N_n stays fixed, and uniqueness for that sign on each vertical line makes

these continuations agree. Every free-boundary point in the smaller cylinder is on one of these graphs.

The graph through p has upward inward normal and will be denoted g_+. If the other orientation occurs, its graph g_- lies below g_+. The vertical rays into the positive sides give $v_j > 0$ above g_+ and below g_- within the smaller cylinder. Between the two graphs there is no further boundary, and just above g_- the value is zero. Thus the whole gap is in the zero phase. If there is just the graph through p, its slope and $g_+(0) = 0$ keep its height within $r/3$ over $|x'| < r$. Each vertical segment below it inside the smaller cylinder is in the zero phase, since it starts on the zero side and meets no other free boundary there. Figure 5 shows the two-sheet geometry.

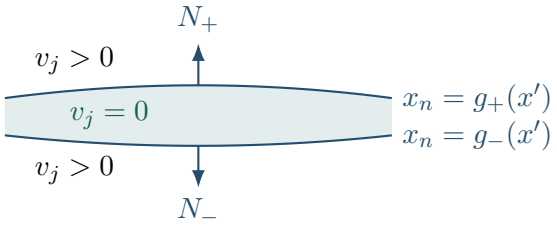


FIGURE 5. A cross-section through two ordered free-boundary sheets. The shaded gap is in the zero phase. The inward normals point into the positive regions on either side.

For derivative bounds, take a point q on one of the graphs over $|x'| < r$, and fix $0 < \rho < r/8$. At any point $x \in B_\rho(q)$, the graph height at x' differs from q_n by at most $\rho/3$. If x is on the inward side, its vertical distance to the graph is at most $4\rho/3 < \ell$, so the ray estimate puts it in the positive phase. Keep v_j on that side in $B_\rho(q)$ and set it to zero on the other side. This is a classical solution in $B_\rho(q)$ with the given sheet as free boundary. In unit-ball coordinates $z = (x - q)/\rho$, divide the solution values by ρ. Its k-th derivatives are then ρ^{k-1} times the corresponding derivatives in the original coordinates. In particular the rescaled Hessian is bounded by ρM. Applying [4, Lemma 3.9] and returning to $B_{\rho/2}(q)$ gives

$$|D^k v_j| \le C_{n,k}\rho^{2-k} M \max\{1, (\rho M)^{k-2}\}, \qquad k \ge 2.$$

These bounds hold on the positive-side traces by continuity.

To obtain graph derivatives, differentiate $v_j(x', g_j(x')) = 0$ in the coordinates of the cylinder. Writing $v_{j,n} = \partial_n v_j$ and evaluating the derivatives of v_j on the graph, we have

$$\partial_i g_j = -\frac{\partial_i v_j}{v_{j,n}}, \qquad \partial_{il} g_j = -\frac{D^2 v_j(e_i + (\partial_i g_j)e_n,\, e_l + (\partial_l g_j)e_n)}{v_{j,n}} \quad (i, l < n).$$

The denominator has absolute value at least $31/32$. For each higher derivative the chain rule again gives $v_{j,n}$ times that derivative of g_j, plus terms involving only lower derivatives of g_j and derivatives of v_j. This gives uniform bounds at all orders over $|x'| < r$ by induction.

Fix $p_\infty \in \Gamma_v$ and choose $p_j \in \Gamma_{v_j}$ tending to it. Center the cylinders at p_j. Take a subsequence on which the coordinate rotations O_j converge to O and the set of graph orientations in the smaller cylinder is fixed. The upper graph always passes through the chart origin. In these coordinates $\widehat{v}_j(x) = v_j(p_j + O_j x)$. Parametrize the positive side of a graph g_j by

$$\Psi_j(x', s) = (x', g_j(x') + \epsilon s), \qquad |x'| < r/2, \quad 0 \le s \le \rho/4,$$

where $\epsilon = 1$ for the upper graph and -1 for the lower. These collars lie on the indicated positive sides for $s > 0$ by the vertical-ray estimate. Each collar point is within $\rho/2$ of the graph point with the same base coordinate. Thus the derivative bounds apply in this fixed parameter domain. The functions themselves are locally bounded by uniform convergence, and their gradients on a

collar are bounded by $1 + Ms$. The bounds on the graphs and on $W_j = \widehat{v}_j \circ \Psi_j$ give smooth convergence on smaller closed collars after a further subsequence, by Arzelà–Ascoli and diagonal extraction over the derivative orders. If Ψ denotes the limiting collar map, the locally uniform convergence identifies the limit of $W_j(x', s)$ with $v(p_\infty + O\Psi(x', s))$.

For small $s > 0$, the bound $\widehat{v}_j(\Psi_j(x', s)) \geq (31/32)s - Ms^2/2$ keeps the limiting collars positive. The locally uniform convergence gives zero on the open gap between the limiting graphs, or on the zero side if only the upper graph is present. In a sufficiently small cylinder about the chart origin these descriptions cover the positive phase. Indeed, for $|x'|, |x_n| < a$ with $a < \rho/16$, the upper graph height is bounded by $a/3$. A positive point above it has vertical distance at most $4a/3$ from it. A positive point below the lower graph has the same bound because $g_- \leq g_+$. The graph and its positive side therefore give a smooth local solution at every point of a limiting sheet where it is separated from the other. At a contact point the two sides are smooth separately. In both cases the positive-side trace of the gradient is the inward unit normal, by the smooth collar convergence and $D\widehat{v}_j = N_j$ on each graph.

The same convergence controls the physical Hessians up to each sheet. The chain rule gives

$$D^2 W_j = (D\Psi_j)^{\mathsf{T}} (D^2 \widehat{v}_j \circ \Psi_j) D\Psi_j + (\partial_n \widehat{v}_j \circ \Psi_j) \begin{pmatrix} D^2 g_j & 0 \\ 0 & 0 \end{pmatrix},$$

$$\partial_n \widehat{v}_j \circ \Psi_j = \epsilon\, \partial_s W_j.$$

Since $D\Psi_j$ is invertible with determinant ϵ, the C^2 convergence of g_j, W_j gives convergence of $D^2\widehat{v}_j$ at convergent collar coordinates, including $s = 0$ by one-sided traces. Convergence of the rotations gives the same conclusion in the original coordinates. These extractions may be made at a single boundary point at a time, since the locally uniform limit v is already fixed.

If there is no contact, v is classical at every free-boundary point and is a global classical stable solution. Suppose instead the two limiting graphs touch. At the contact the function $g_+ - g_- \geq 0$ has a zero minimum. Hence $Dg_+ = Dg_-$ and $D^2(g_+ - g_-)$ is positive semidefinite there. With $F(z) = z/\sqrt{1 + |z|^2}$, the inward mean curvatures are $H_+ = -\operatorname{div} F(Dg_+)$ and $H_- = \operatorname{div} F(Dg_-)$. To compute them, extend the upward unit normal $(-Dg, 1)/\sqrt{1 + |Dg|^2}$ independently of x_n. Since it has unit length, its derivative in its own direction has zero normal component, so its ambient divergence equals its tangential divergence on the graph. The lower sheet uses the downward normal. Thus at contact

$$H_+ + H_- = -DF(Dg_+) : D^2(g_+ - g_-) \leq 0.$$

The sign follows because $DF(z)$ is positive definite:

$$\xi^{\mathsf{T}} DF(z)\xi = \frac{(1 + |z|^2)|\xi|^2 - (z \cdot \xi)^2}{(1 + |z|^2)^{3/2}} \geq \frac{|\xi|^2}{(1 + |z|^2)^{3/2}}.$$

On each smooth positive side, $\sigma = 1 - |Dv|^2 \geq 0$ since v is 1-Lipschitz, and $\sigma = 0$ at its boundary. The identities (3.1) on that side give

$$\partial_{N_\pm} \sigma = 2H_\pm, \qquad \Delta\sigma = -2|D^2 v|^2 \leq 0.$$

The inward derivatives of σ at contact are nonnegative, so the inequality for $H_+ + H_-$ forces both to vanish. In a small connected domain on either positive side, the strong minimum principle and the Hopf lemma at the smooth boundary point give a strictly positive inward derivative unless σ is identically zero there. Thus $\sigma = 0$ on both sides locally and the Laplacian identity gives $D^2 v = 0$ there.

Harmonic unique continuation extends each local affine formula to its entire positive component. The two gradients are opposite unit vectors, so the components are distinct. The relative open-and-closed argument of Section 5 fills the corresponding positive half-spaces: on the boundary of each component its affine formula vanishes by continuity. The half-spaces are complementary

across their common boundary plane through the contact point. Hence v is the absolute value of the affine function with unit gradient defining one half-space. Since $0 \in \Gamma_v$, this affine function vanishes at the origin.

Proof of Corollary 1.4. In dimension one $u'' = 0$ by harmonicity, and the curvature bound is vacuous. In higher dimensions the Bernoulli rescaling $u_R(y) = R^{-1}u(x_0 + Ry)$ preserves the gradient at corresponding points and multiplies the Hessian and free-boundary curvatures by R. The stability-form scaling from Section 4 in dimension n is $Q_{u_R}(\phi) = R^{2-n}Q_u(\phi((\cdot - x_0)/R))$. Here the squared-gradient and volume factors are R^2, R^{-n}, while the curvature and surface factors are $R, R^{-(n-1)}$. Thus the rescaled solution is stable by Appendix A. It suffices to take $R = 1$, $x_0 = 0$. If the Hessian bound failed, choose classical stable u_j in B_2, with $0 \in \Gamma_{u_j}$, such that $\sup_{\Omega_j \cap B_1} |D^2 u_j| \to \infty$, where $\Omega_j = \{u_j > 0\}$. On $\overline{B_{5/4}} \cap \overline{\Omega_j}$, maximize $(5/4 - |x|)|D^2 u_j(x)|$, using one-sided boundary traces. Let p_j be a maximizing point, $d_j = |D^2 u_j(p_j)|$, and $\rho_j = 5/4 - |p_j|$. The weighted Hessian is continuous up to the free boundary from the positive phase, so the maximum exists. It is positive for large j, hence $\rho_j > 0$. The weight is at least $1/4$ on B_1, so the maximum of the weighted Hessian tends to infinity. On $B_{\rho_j/2}(p_j)$ the weight is at least $\rho_j/2$, so maximality also bounds the Hessian there. In formulas,

$$m_j = d_j \rho_j \longrightarrow \infty, \qquad |D^2 u_j| \le 2 d_j \quad \text{in } B_{\rho_j/2}(p_j) \cap \Omega_j.$$

We keep a free-boundary point at bounded distance from the Hessian normalization after rescaling. If p_j is in the positive phase, set $s_j = \min\{\operatorname{dist}(p_j, \Gamma_{u_j}), \rho_j/4\}$. The ball $B_{s_j}(p_j)$ is positive and lies in $B_{3/2}$, since $|p_j| + \rho_j/4 \le 5/4$. Applying the interior gradient estimate to the harmonic components of Du_j there gives $s_j d_j \le C_n L_n$ by (6.1). Since $d_j \rho_j \to \infty$, for large j we must have $s_j < \rho_j/4$. The distance to Γ_{u_j} is then attained at a point z_j in the compact ball $\overline{B_{\rho_j/4}(p_j)} \subset B_2$. Hence $d_j |p_j - z_j| \le A_n$ for a dimensional A_n. If $p_j \in \Gamma_{u_j}$, take $z_j = p_j$, for which the same bound holds.

Define $v_j(y) = d_j u_j(z_j + y/d_j)$ and $q_j = d_j(p_j - z_j)$. The scaling preserves stability and gives $D^2 v_j(y) = d_j^{-1} D^2 u_j(z_j + y/d_j)$. In these coordinates the ball $B_{\rho_j/2}(p_j)$ becomes $B_{m_j/2}(q_j)$. Since $|q_j| \le A_n$, it contains $B_{m_j/2 - A_n}$. Thus $0 \in \Gamma_{v_j}$, $|D^2 v_j| \le 2$ on the positive phase in $B_{m_j/2 - A_n}$, and $|D^2 v_j(q_j)| = 1$, with the one-sided trace if needed. These are the hypotheses of the compactness argument above. The classical limit is flat by Theorem 1.1 or Corollary 1.3, and the absolute-value limit is flat on both positive sides. Pass to a subsequence with $q_j \to q$. The local Hausdorff convergence of the closed positive phases and $q_j \in \overline{\{v_j > 0\}}$ give $q \in \overline{\{v > 0\}}$. If $v(q) > 0$, interior harmonic estimates give $D^2 v_j(q_j) \to D^2 v(q) = 0$.

If $q \in \Gamma_v$, take the moving charts above centered at boundary points tending to q. For large j, q_j is in one of the closed positive-side collars in that chart. After passing to a subsequence use a fixed side and write its collar coordinates as (x_j', s_j) with $s_j \ge 0$. The base coordinates tend to zero. The ray lower bound $(31/32)s_j - M s_j^2/2$ and $v_j(q_j) \to v(q) = 0$ give $s_j \to 0$. The Hessian convergence in the collars therefore gives $D^2 v_j(q_j) \to 0$ on the chosen side, including the trace if $s_j = 0$. In either case this contradicts $|D^2 v_j(q_j)| = 1$.

For tangent vectors X, Y on Γ, differentiating $Du = N$ gives $D^2 u(X, Y) = \langle D_X N, Y \rangle = \mathrm{II}_\Gamma(X, Y)$ with the inward sign convention. Thus the tangential block of the positive-side Hessian is II_Γ, and $|\mathrm{II}_\Gamma| \le |D^2 u|$ on the trace. The interior bound passes to this trace by continuity and gives the second inequality in (1.3). □

6.2. **Finite-index solutions.**

Proof of Corollary 1.5. In [8, (1.2)], the curvature sign is $H = -\operatorname{div}_\Gamma \nu$ for the outward normal ν. This equals our $\operatorname{div}_\Gamma N$. Their tests are smooth on the closed positive side with compact support in the ambient domain. Such tests extend smoothly across the boundary by flattening

it in finitely many charts near the support, extending from the half-space, and patching with compactly supported cutoffs and a partition of unity. Ambient tests also restrict to positive-side tests. Both operations preserve Q, including its boundary term.

Restriction is injective on a negative subspace of ambient tests, since a test with zero restriction also has zero trace and $Q = 0$. Conversely, extend a basis of any finite-dimensional negative subspace of positive-side tests. Restricting the extensions recovers that basis, so their span has the same dimension and is again negative definite. The two indices therefore agree. The same correspondence matches scalar stability on the whole space or on the punctured space, using compact supports in the respective ambient domain.

The classification applies for $3 \le n < \min(n_*^1, n_*^2)$, where n_*^1, n_*^2 are the critical dimensions in [8, Theorem 1.2 and (1.3)–(1.5)]. The first is the least dimension with a non-one-dimensional entire classical stable solution. Scalar stability gives inner stability by Appendix A. Theorem 1.1 and Corollary 1.3 therefore give $n_*^1 \ge 7$.

The second critical dimension concerns one-homogeneous stable solutions that are classical in the punctured space. Such a solution extends continuously at the origin by zero, since $u(r\theta) = ru(\theta)$ and u is bounded on the unit sphere. Scalar stability gives inner stability for deformations supported away from the origin by the same Appendix A calculation. The cone exclusions in Section 2 apply in dimensions two through six. Dimension one already has a single coordinate, so $n_*^2 \ge 7$. Both dimension hypotheses hold for the range of the corollary.

For a solution depending on one linear coordinate, rotations about that coordinate axis fix u. For any other solution, [8, Theorem 1.2] gives the two alternatives, the monotonicity, and the zero set in the statement. The exterior-ball solution is invariant under rotations about every line through its center.

In the graph alternative, g is differentiable on $(1, \infty)$ with $g'(r) > 0$ there. In particular $g(r) > 0$ for $r > 1$, by the mean value theorem and $g \ge 0$. For $|\omega| = 1$, the upper free boundary at $(r\omega, g(r))$ contains the radial tangent $(\omega, g'(r))$ and sphere tangents $(\tau, 0)$ for $\tau \perp \omega$. Their horizontal projections span $\mathbb{R}^{n-1}$, so the horizontal projection restricted to the smooth free boundary has invertible derivative. The inverse function theorem makes the upper height smooth near $r\omega$, hence g is smooth for $r > 1$. Monotonicity gives a right limit at 1. If this limit were positive, choose $s > 0$ below it. The points $(r\omega, s)$ would be in the zero set for $r > 1$ close to 1. By continuity so would (ω, s), contrary to the zero-set formula and $g(1) = 0$. Thus g is continuous at 1. The finite limit at infinity from [8, Theorem 1.2] is at least $g(2) > 0$. □

7. Hessian energy and neck exclusion

In dimension six, the integral of $|D^2u|^6$ is invariant under Bernoulli scaling. Marked necks carry a fixed positive amount of this energy. We bound it on a smaller ball by combining height closeness to a two-sided profile on a larger ball with a Hessian bound on the smaller ball.

Proposition 7.1 (Hessian energy at small height excess)**.** *Let u be classical stable in $B_{8R}(z_0) \subset \mathbb{R}^6$, with $|Du| \le 1$ in the positive phase there. For a unit vector e, set $V(x) = |e \cdot (x - z_0)|$ and*

$$\varepsilon = \left(R^{-8} \int_{B_{8R}(z_0)} (u - V)^2 \right)^{1/2}.$$

For sufficiently small ε, every ball $B_{2r}(z) \subseteq B_{2R}(z_0)$ satisfies

$$\int_{\Omega \cap B_r(z)} |D^2u|^2 \le C\varepsilon^{1/4} r^4. \tag{7.1}$$

If also $\sup_{\Omega\cap B_r(z)}|D^2u|\le K/r$, *then*

$$\int_{\Omega\cap B_r(z)}|D^2u|^6\le CK^4\varepsilon^{1/4}. \tag{7.2}$$

The smallness threshold and C *are dimensional.*

The distributional Laplacian records the area of the free boundary. For a smooth test χ compactly supported in the ambient domain, harmonicity gives $\operatorname{div}(\chi Du)=Du\cdot D\chi$ in the positive phase. The only boundary flux is on Γ, where $\chi\partial_\nu u=-\chi$ for the outward normal ν. Thus

$$\int_\Gamma\chi=-\int_\Omega Du\cdot D\chi. \tag{7.3}$$

By the weak-gradient formula in Section 1, the right-hand side equals $\langle\Delta u,\chi\rangle$. Thus, in dimension six, $\Delta u=\mathcal{H}^5\lfloor\Gamma$.

Proof of Proposition 7.1. Set $x=z_0+Ry$ and $\widehat{u}(y)=R^{-1}u(x)$. The rescaled comparison profile is $|e\cdot y|$. Since $dx=R^6\,dy$ and $u(x)-V(x)=R(\widehat{u}(y)-|e\cdot y|)$,

$$\varepsilon^2=\int_{B_8}(\widehat{u}(y)-|e\cdot y|)^2\,dy.$$

The gradient and stability bounds are preserved by the rescaling in the proof of Theorem 1.2. Also $B_{2r}(z)$ becomes $B_{2r/R}((z-z_0)/R)\subseteq B_2$. Work first in the normalized coordinates, using u,V,z,r for the normalized data. Write $v=u-V$ and $w=|D^2u|$ in the positive phase for the Hessian norm.

By the weak-gradient formula in Section 1 and the hypothesis $|Du|\le1$ in the positive phase, the ambient weak gradient of u has norm at most one almost everywhere. This gives the 1-Lipschitz bound on the convex ball B_8. Mollify on a neighborhood of any segment compactly contained in B_8, integrate the gradient bound along the segment, and pass to the locally uniform limit. As V is 1-Lipschitz, v is 2-Lipschitz. Fix $x\in B_6$ with $m=|v(x)|>0$, and take $s=\min\{1,m/4\}$. For $y\in B_s(x)$, $|v(y)|\ge m-2s\ge m/2$. Since $B_s(x)\subset B_8$,

$$\varepsilon^2\ge\int_{B_s(x)}|v|^2\ge\tfrac14|B_1|\,m^2s^6.$$

For small ε, $m\ge4$ is impossible because it would give $s=1$ and $\varepsilon^2\ge4|B_1|$. Using $s=m/4$ now gives $m^8\le C\varepsilon^2$. Thus

$$\|v\|_{L^\infty(B_6)}\le C\varepsilon^{1/4}.$$

To control the gradient error, first bound the measure $\mu=\Delta v$ on B_5. For a nonnegative cutoff $\vartheta\in C_c^\infty(B_6)$ equal to one on B_5, (7.3) gives

$$\Delta u(B_5)\le\int_\Gamma\vartheta=-\int_\Omega Du\cdot D\vartheta\le\int|D\vartheta|.$$

For $x=y+te$ with $y\in e^\perp$, the weak gradient of V is $\operatorname{sgn}(t)e$ almost everywhere. For a smooth compactly supported test ϕ, integrating $-\operatorname{sgn}(t)\partial_t\phi(y+te)$ over the two half-lines gives $2\phi(y)$. Integration in y yields $\Delta V=2\mathcal{H}^5\lfloor e^\perp$. The second measure has mass $2\mathcal{H}^5(e^\perp\cap B_5)$ on B_5. Since both measures are nonnegative, their sum bounds $|\mu|$ there and gives $|\mu|(B_5)\le C$.

Choose $0\le\chi\le1$ smooth, equal to one on B_4, and compactly supported in B_5. We test $\mu=\Delta v$ with χ^2v, as in the gradient estimate in the proof of [9, Proposition 4.12]. The test is Lipschitz with compact support in B_5. Mollifying it gives smooth tests supported in B_5 that

converge uniformly and in H^1. These two convergences pass to the integrals against μ and Dv, respectively. Expanding the gradient of the test and using $D(v^2) = 2vDv$ then gives

$$\begin{aligned}\int \chi^2 |Dv|^2 &= -\int \chi^2 v \, d\mu - \tfrac{1}{2}\int D(\chi^2)\cdot D(v^2) \\ &= -\int \chi^2 v \, d\mu + \tfrac{1}{2}\int v^2 \Delta(\chi^2).\end{aligned}$$

The absolute value of the measure term is at most $\|v\|_{L^\infty(B_6)}|\mu|(B_5) \le C\varepsilon^{1/4}$, and the other term is at most $C\|v\|^2_{L^2(B_8)} = C\varepsilon^2$. Since $\chi = 1$ on B_4, for $\varepsilon \le 1$ this proves $\int_{B_4} |Dv|^2 \le C\varepsilon^{1/4}$.

On Ω, the gradient comparison of Section 1.4 applies with $q = DV = \pm e$ almost everywhere and gives $h_e^2 \le (1 + 4/30)|Dv|^2$. Thus the defect on B_4 satisfies

$$E_e(B_4)^2 = 4^{-6}\int_{\Omega\cap B_4} h_e^2 \le C\varepsilon^{1/4}.$$

In particular $E_e(B_4) \le C\varepsilon^{1/8}$. Choose the height threshold small enough that this is at most ε_0 in Theorem 1.2. Applying that theorem on B_4 and squaring its bound gives

$$\sup_{\Omega\cap B_2} h_e^2 \le C\varepsilon^{1/4}.$$

For $B_{2r}(z) \subseteq B_2$, take $\eta = 1$ on $B_r(z)$, smooth and compactly supported in $B_{2r}(z)$, with $|D\eta| \le C/r$. Lemma 3.1 gives the Robin inequality for h_δ and $h_\delta \Delta h_\delta \ge w^2/75$. Thus (4.1) with $G = h_\delta$ yields

$$\tfrac{1}{75}\int_{\Omega\cap B_r(z)} w^2 \le \int_\Omega \eta^2 h_\delta \Delta h_\delta \le \int_\Omega (h_e^2 + \delta^2)|D\eta|^2.$$

The term with δ^2 vanishes as $\delta \downarrow 0$. The remaining integral is at most $C\varepsilon^{1/4} r^{-2}|B_{2r}(z)| \le C\varepsilon^{1/4} r^4$, since $|B_{2r}(z)| = |B_1|(2r)^6$.

In the original coordinates, $D^2\widehat{u}(y) = RD^2u(z_0 + Ry)$ and $dy = R^{-6}\,dx$. With $\widehat{z} = (z - z_0)/R$, this gives

$$\int_{\{\widehat{u}>0\}\cap B_{r/R}(\widehat{z})} |D^2\widehat{u}|^2 = R^{-4}\int_{\Omega\cap B_r(z)} |D^2u|^2.$$

The normalized bound has radius factor $(r/R)^4$, so multiplying by R^4 gives (7.1). If $w \le K/r$ on the original positive ball, then

$$\int_{\Omega\cap B_r(z)} w^6 \le (K/r)^4 \int_{\Omega\cap B_r(z)} w^2 \le CK^4\varepsilon^{1/4},$$

which is (7.2). □

For the neck application, consider the normalized contradiction setting of [9, Assumption 5.3] in $n = 6 < n_*$, where the dimension bound is from Section 2. In this setting $|Du| \le 1$ and $|D^2u| \le 1$ in Ω. Let $\mathcal{Z}$ be the marked centers of [9, Definition 5.4], and let $\eta_0 > 0$ be the dimensional threshold in [9, (5.2)]. For a marked center z, set

$$I_z(s) = \int_{\Omega\cap B_s(z)} |D^2u|^6, \qquad r_*(z) = \inf\{s > 0 : I_z(s) \ge \eta_0^6\}.$$

The bound $I_z(s) \le |B_1|s^6$ gives $r_*(z) > 0$. Marked centers are selected only from the finite-radius classes in [9, Definition 5.4], so $r_*(z) < \infty$. The integral $I_z(s)$ is continuous in $s > 0$: the ball indicators converge away from the limiting sphere, which has Lebesgue measure zero, and the integrands are bounded by one on a fixed larger ball. It is less than η_0^6 below $r_*(z)$, and radii with $I_z(s) \ge \eta_0^6$ approach $r_*(z)$. Hence $I_z(r_*(z)) = \eta_0^6$. Lemma 5.7 of [9], with $M = 1$, gives $|D^2u| \le K/r_*(z)$ on $\Omega \cap B_{r_*(z)}(z)$ for a dimensional $K > 0$.

For a direction e at scale $8R$, write $M^e_{8R,z_0} = (8R)^{-8}\int_{B_{8R}(z_0)}(u-V)^2$ for the height term in [9, (7.8)]. The square of the combined excess $\widetilde{\boldsymbol{E}}_{z_0}(8R;e)$ is the sum of this term and the nonnegative Weiss deficit contribution from [9, Lemma 7.4]. Consequently

$$\varepsilon = 8^4\sqrt{M^e_{8R,z_0}} \le 8^4\widetilde{\boldsymbol{E}}_{z_0}(8R;e).$$

For a marked z with $B_{2r_*(z)}(z) \subseteq B_{2R}(z_0)$, the Hessian bound and (7.2) from Proposition 7.1 give

$$\eta_0^6 = I_z(r_*(z)) \le CK^4\varepsilon^{1/4}.$$

Let $\bar\varepsilon$ be the height threshold in Proposition 7.1. If the minimized combined excess at $8R$ is less than

$$8^{-4}\min\left\{\bar\varepsilon, \left(\frac{\eta_0^6}{CK^4}\right)^4\right\},$$

take a direction with combined excess below this bound. Then $\varepsilon < \bar\varepsilon$ and $CK^4\varepsilon^{1/4} < \eta_0^6$, excluding every marked ball satisfying the containment. In particular the containment is automatic for $z = z_0 \in \mathcal{Z}$ with $r_*(z_0) \le R$.

Appendix A. Scalar and inner stability

Let u be classical in an open set A, with the normal and curvature conventions of Section 1. We will use smooth functions and vector fields on the closed positive side with compact support in A. They extend smoothly across Γ by the boundary-chart extension used for the test spaces in Section 6.2. The extension can be kept compactly supported in A.

On Γ, differentiating $Du = N$ along any tangent vector τ gives $D^2u\,\tau = D_\tau N$. Since $|N| = 1$, $D_\tau N$ is tangent, so the mixed entries of the Hessian vanish by symmetry. For an orthonormal tangent basis (τ_i), the trace identity is

$$0 = \Delta u = D^2u(N,N) + \sum_i \langle D_{\tau_i}N, \tau_i\rangle = D^2u(N,N) + H.$$

Consequently $D^2u\,N = -HN$.

Let Φ_t be a smooth family of diffeomorphisms with $\Phi_0 = \mathrm{id}$, equal to the identity outside a fixed compact subset of $D \Subset A$. Set $u_t = u\circ\Phi_t^{-1}$, $\Omega_t = \Phi_t(\Omega)$, $\Gamma_t = \Phi_t(\Gamma)$, and $J(t) = J(u_t;D)$. The velocity at the moving spatial point is $X_t = \dot\Phi_t\circ\Phi_t^{-1}$, and the outward normal speed of Γ_t is $V_t = X_t\cdot\nu_t$. Differentiating $u_t\circ\Phi_t = u$ gives $\dot u_t = -Du_t\cdot X_t$ in Ω_t, where the dot on u_t denotes differentiation at a fixed spatial point. On Γ_t we take its positive-side trace. There $Du_t = -|Du_t|\nu_t$, since u_t vanishes on Γ_t and is positive on the inward side. Thus $\dot u_t = |Du_t|V_t$ on Γ_t. At $t=0$ we write $X = X_0$, $\psi = \dot u_0 = -Du\cdot X$, so $V_0 = \psi$ on Γ.

Since Φ_t fixes the complement of D, it maps D onto itself. Pull the energy integral over $\Omega_t\cap D$ back to $\Omega\cap D$. Differentiating $\dot\Phi_t = X_t\circ\Phi_t$ in space gives $D\dot\Phi_t = (DX_t\circ\Phi_t)D\Phi_t$. The determinant derivative therefore gives $\dot j_t = (\operatorname{div}X_t)\circ\Phi_t\, j_t$ for $j_t = \det D\Phi_t$. The chain rule in the pulled-back integrand and the Jacobian derivative give, in moving coordinates,

$$J'(t) = \int_{\Omega_t\cap D}\big(2Du_t\cdot D\dot u_t + \operatorname{div}\big((1+|Du_t|^2)X_t\big)\big).$$

The flux fields $(1+|Du_t|^2)X_t$ and $\dot u_t Du_t$ vanish near ∂D, so only the free boundary contributes. The divergence theorem and Green's identity give

$$\begin{aligned} J'(t) &= 2\int_{\Omega_t\cap D} Du_t\cdot D\dot u_t + \int_{\Gamma_t\cap D}(1+|Du_t|^2)V_t \\ &= -2\int_{\Omega_t\cap D}\Delta u_t\,\dot u_t + \int_{\Gamma_t\cap D}(1-|Du_t|^2)V_t. \end{aligned}$$

Indeed, the boundary term from the Dirichlet integral is $2\dot{u}_t\partial_{\nu_t}u_t = -2|Du_t|^2V_t$.

To differentiate $J'(t)$ at zero, pull the two integrals back by Φ_t. At $t = 0$ the factors Δu in the bulk and $1 - |Du|^2$ on the boundary vanish. Hence derivatives of the velocities and of the volume and surface Jacobians drop out. In the bulk the derivative of $\Delta u_t \circ \Phi_t$ at zero is $\Delta\psi + X \cdot D(\Delta u) = \Delta\psi$. The material derivative of Du_t at zero is $D\psi + D^2u\,X$ by the chain rule in the interior. The formula extends to its positive-side trace on Γ. Hence the boundary coefficient has derivative

$$-2N \cdot (D\psi + D^2u\,X) = 2(\partial_\nu\psi - H\psi).$$

Here $-N = \nu$ and $N \cdot D^2u\,X = (D^2u\,N) \cdot X = -HN \cdot X = H\psi$. Using $V_0 = \psi$ and integrating by parts once more yields

$$J''(0) = -2\int_\Omega \psi\,\Delta\psi + 2\int_\Gamma \psi\,\partial_\nu\psi - 2\int_\Gamma H\psi^2 = 2Q(\psi).$$

The integrands have compact support in D. Since ψ extends to a smooth compactly supported ambient test, (1.2) implies inner stability.

Conversely, suppose u is inner stable and fix a signed $\phi \in C_c^\infty(A)$. Where $Du \neq 0$ in the positive phase, the choice $X = -\phi Du/|Du|^2$ gives $-Du \cdot X = \phi$. To use this construction on the entire support, we remove the interior critical set by cutoffs. The argument may be made on each connected component of A, since the terms of Q add. If Γ is empty on a component, its contribution to Q is already nonnegative. On the remaining components, u is nonconstant on each positive component. Indeed, a relative boundary point of a positive component has value zero, as in Section 5. A component on which u is a positive constant therefore has no such boundary point and fills its ambient component. In dimension one the scalar inequality holds directly because the tangential trace H is zero, so we can take $n \geq 2$ below.

The critical set $\mathcal{C} = \{Du = 0\} \cap \Omega \cap \operatorname{supp}\phi$ on the components under consideration is compactly contained in Ω. Indeed, a limit on Γ would contradict the continuous positive-side gradient trace with $|Du| = 1$. Stratify $\mathcal{C}$ by the least order $m \geq 2$ for which a derivative of u is nonzero at the point. Such an order is finite by real analyticity of nonconstant harmonic functions. At a point of order m choose an order-$(m-2)$ derivative f of u for which $D^2f \neq 0$. This Hessian is trace free by harmonicity and thus has rank at least two. Two components of Df therefore have linearly independent gradients at the point. These components are derivatives of u of order $m - 1$, so they vanish at every point in the same stratum. The implicit function theorem puts that stratum locally in a smooth submanifold of codimension two. Taking a countable chart cover in each stratum and compact pieces of the charts covers $\mathcal{C}$.

Each compact chart piece admits a smooth cutoff equal to one in a neighborhood of the piece and with arbitrarily small H^1 norm, supported inside Ω. In flattened coordinates, multiply a tangential cutoff by a radial cutoff in the two normal variables. For normal radius r, let it be one for $r \leq \epsilon^2$, zero for $r \geq \epsilon$, and a smooth function of $\log(\epsilon/r)/|\log\epsilon|$ in between. Its derivative is bounded by $C/(r|\log\epsilon|)$. The squared L^2 norm and the tangential derivative cost are $O(\epsilon^2)$. The normal derivative cost is bounded by

$$\frac{C}{|\log\epsilon|^2}\int_{\epsilon^2}^{\epsilon}\frac{dr}{r} = \frac{C}{|\log\epsilon|}.$$

Changes of coordinates on the compact support preserve these bounds up to fixed constants. For each j, choose the cutoff for chart piece i with H^1 norm at most $2^{-i}/j$. The neighborhoods where the cutoffs equal one cover $\mathcal{C}$, so finitely many suffice. Sum that finite collection and compose with a smooth function that is zero at zero, equals one on $[1, \infty)$, and has bounded derivative and values in $[0, 1]$. This gives $\eta_j \in C_c^\infty(\Omega)$ equal to one near $\mathcal{C}$, with $\|\eta_j\|_{H^1} \leq C\sum_{i\geq 1} 2^{-i}/j = C/j$ by the triangle inequality and the chain rule.

Set $\phi_j = \phi(1-\eta_j)$. Its trace on Γ is ϕ, and $\phi_j \to \phi$ in H^1. The field $X_j = -\phi_j Du/|Du|^2$ is smooth on the positive side where $Du \neq 0$ and extends by zero near the critical points, since ϕ_j vanishes there. Extend X_j smoothly and with compact support across Γ as above. Its flow realizes $\psi = \phi_j$ in the second-variation identity, so $Q(\phi_j) \geq 0$. The bulk term converges by the H^1 convergence and the boundary term is unchanged along this sequence. Thus $Q(\phi) \geq 0$ for all signed tests.

Appendix B. Weighted local curvature estimate

The local Hessian bound also yields a weighted estimate for third derivatives. A regularized power of the Hessian norm controls these derivatives through its Laplacian, and the Bernoulli condition bounds its normal derivative on the free boundary.

Theorem B.1 (Local curvature growth)**.** *Let u be classical stable in $B_{4R}(x_0) \subset \mathbb{R}^6$, with $x_0 \in \Gamma$, and write $w = |D^2 u|$. Then*

$$\int_{\Omega \cap B_R(x_0)} (w^{42/11} + w^{-2/11}|D^3 u|^2) + \int_{\Gamma \cap B_R(x_0)} w^{31/11} \leq C R^{24/11}. \tag{B.1}$$

The constant C is dimensional. The weighted integrand is defined to be zero where $w = 0$, and the boundary integral uses the one-sided trace.

Proof of Theorem B.1. Write $B_s = B_s(x_0)$ throughout the proof. Corollary 1.4 applied with radius $2R$ gives $w \leq C/R$ on $\Omega \cap B_{2R}$. The same bound holds on $\Gamma \cap B_{2R}$ by continuity from the positive side. The rescaled function $v(y) = (2R)^{-1} u(x_0 + 2Ry)$ is classical in the ball of radius two centered at zero, with $0 \in \Gamma_v$ and $Dv(y) = Du(x_0 + 2Ry)$. Thus (6.1) gives $|Du| \leq C$ in $\Omega \cap B_{3R}$. Choose a smooth cutoff $0 \leq \chi \leq 1$ equal to one on B_{2R} and compactly supported in B_{3R}, with $|D\chi| \leq C/R$. By (7.3),

$$\mathcal{H}^5(\Gamma \cap B_{2R}) \leq \int_\Gamma \chi = -\int_\Omega Du \cdot D\chi \leq C R^{-1} |B_{3R}| \leq C R^5.$$

Consequently

$$\int_{\Omega \cap B_R} w^{42/11} \leq (C/R)^{42/11} |B_R| \leq C R^{6-42/11} = C R^{24/11},$$
$$\int_{\Gamma \cap B_R} w^{31/11} \leq (C/R)^{31/11} \mathcal{H}^5(\Gamma \cap B_{2R}) \leq C R^{5-31/11} = C R^{24/11}.$$

For the remaining term, set $T = D^2 u$, $\beta = 10/11$, $W_\delta = w^2 + \delta^2$, and $f_\delta = W_\delta^\beta$, where $0 < \delta \leq R^{-1}$. The components $T_{jk} = \partial_j \partial_k u$ satisfy $\Delta T_{jk} = \partial_j \partial_k \Delta u = 0$. Differentiating $W_\delta = \sum_{j,k} T_{jk}^2 + \delta^2$ gives

$$\partial_i W_\delta = 2 \sum_{j,k} T_{jk} \partial_i T_{jk}, \qquad \Delta W_\delta = 2 \sum_{i,j,k} (\partial_i T_{jk})^2 = 2|DT|^2.$$

By Cauchy–Schwarz, $|DW_\delta|^2 \leq 4w^2 |DT|^2$. The gradient term in the chain rule has coefficient $\beta(\beta-1) < 0$, so this estimate gives

$$\begin{aligned} \Delta f_\delta &= \beta W_\delta^{\beta-1} \Delta W_\delta + \beta(\beta-1) W_\delta^{\beta-2} |DW_\delta|^2 \\ &\geq 2\beta W_\delta^{\beta-2} (\delta^2 + (2\beta-1) w^2) |DT|^2 \\ &\geq 2\beta(2\beta-1) W_\delta^{\beta-1} |DT|^2. \end{aligned}$$

The last inequality follows from $0 < 2\beta - 1 < 1$.

To estimate the boundary flux, let S be the tangential block of T. By Appendix A, $D_X N = SX$ for tangent X and $T\nu = -H\nu = HN$. Since $\operatorname{tr} T = 0$, we have $\operatorname{tr} S = H$. The mixed entries of T

vanish, so $w^2 = |S|^2 + H^2$. Differentiating $T\nu = HN$ along a tangent vector X and pairing with a tangent vector Y gives

$$D^3u(X,\nu,Y) + \langle TD_X\nu, Y\rangle = (XH)\langle N,Y\rangle + H\langle D_XN, Y\rangle.$$

Using $D_X\nu = -SX$ and $\langle N,Y\rangle = 0$, we obtain $D^3u(\nu,X,Y) = \langle (S^2+HS)X,Y\rangle$. For an orthonormal tangent basis $(\tau_i)_{i=1}^5$, the positive-side trace of $\partial_\nu\Delta u = 0$ gives

$$D^3u(\nu,\nu,\nu) = -\sum_{i=1}^5 D^3u(\nu,\tau_i,\tau_i) = -|S|^2 - H\operatorname{tr}S = -w^2.$$

At the point of evaluation choose a fixed Euclidean orthonormal frame $(\tau_1,\dots,\tau_5,\nu)$. In this frame T has blocks S and $-H$, so differentiating the squared norm gives

$$\tfrac12\partial_\nu(w^2) = \operatorname{tr}\big(S(S^2+HS)\big) + (-H)(-w^2) = \operatorname{tr}(S^3) + H|S|^2 + Hw^2.$$

If λ_i are the eigenvalues of S, then $|\operatorname{tr}(S^3)| \le \sum_i |\lambda_i|^3 \le |S|^3$. As $|S|, |H| \le w$, this gives $|\partial_\nu(w^2)| \le 6w^3$. For $w > 0$, $W_\delta \ge w^2$ and $\beta - 1 < 0$, so

$$|\partial_\nu f_\delta| \le 6\beta W_\delta^{\beta-1}w^3 \le 6\beta w^{2\beta+1}.$$

At $w = 0$, the bound $|\partial_\nu(w^2)| \le 6w^3$ gives $\partial_\nu f_\delta = 0$.

Choose a smooth cutoff $0 \le \kappa \le 1$ equal to one on B_R and compactly supported in B_{2R}. Scaling a fixed cutoff gives $|D\kappa| \le C/R$ and $|\Delta\kappa| \le C/R^2$. The divergence of $\kappa Df_\delta - f_\delta D\kappa$ is $\kappa\Delta f_\delta - f_\delta\Delta\kappa$. The field is supported away from ∂B_{4R}, so the divergence theorem gives

$$\int_\Omega \kappa\Delta f_\delta = \int_\Omega f_\delta\Delta\kappa + \int_\Gamma \kappa\partial_\nu f_\delta - \int_\Gamma f_\delta\partial_\nu\kappa.$$

On the support of the cutoff, $w \le C/R$ and $\delta \le R^{-1}$ give $f_\delta \le CR^{-2\beta}$ and $|\partial_\nu f_\delta| \le CR^{-(2\beta+1)}$ on Γ. The absolute values of the three terms on the right are bounded by $CR^{6-2\beta-2}$, $CR^{5-(2\beta+1)}$, and $CR^{5-2\beta-1}$, respectively, using the volume and area bounds for B_{2R}. All three exponents are $4 - 2\beta = 24/11$. Since $\kappa \ge 0$ and $\kappa = 1$ on B_R, the lower bound for Δf_δ yields

$$2\beta(2\beta-1)\int_{\Omega\cap B_R} W_\delta^{\beta-1}|DT|^2 \le \int_\Omega \kappa\Delta f_\delta \le CR^{24/11}.$$

On $\Omega \cap B_R \cap \{w = 0\}$, the regularized weight equals $\delta^{-2/11}$. Thus the integral of $|DT|^2$ over this set is at most $CR^{24/11}\delta^{2/11}$ for every such δ, and $DT = 0$ there almost everywhere. On $\{w > 0\}$, the weight increases to $w^{-2/11}$ as $\delta \downarrow 0$. Monotone convergence now gives

$$\int_{\Omega\cap B_R} w^{-2/11}|D^3u|^2 \le CR^{24/11}$$

with the stated zero-set convention. Adding the two bounds for the powers of w proves (B.1). □

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